

Ahaan Sandeep Sarkar | Kandidatnr: 1005  
Lukas Nissen Molvær | Kandidatnr: 201

Emnekode: BAO302

Emnenavn: Bachelorprosjekt

Oppdragsgiver: AgroIntel (Kristian Vastveit)

Innleveringsdato: 14.05.2025 Antall sider: 84

Antall ord: 12593

Tilgjengelighet: Fri ☐ Begrenset ☒

Høgskolen Kristiania

KI-basert storfeovervåkingssystem for "AgroIntel"

AI-Based Cattle Monitoring System For 'AgroIntel'



Semester: Våren 2025

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# Bachelor Project Report: AI-Based Cattle Monitoring System

Lukas N. Molvær  
Ahaan R. S. Sarkar

Høyskolen Kristiania  
In collaboration with



# AGROINTEL

May 14, 2025

## 0.1 Acknowledgments

We would like to express our sincere gratitude to our academic supervisor, Mr. Arvind Keprate, for his continued guidance, constructive feedback, and encouragement throughout the course of this project. His support helped us navigate the technical and design challenges effectively.

We are also thankful to the team at AgroIntel, especially our founder Mr. Kristian Vastveit, for offering valuable domain insight and for providing us with the opportunity to collaborate with a forward-thinking organization at the intersection of agriculture and technology, as well as for providing the resources and facilities necessary for our work.

Finally, we acknowledge the developers and contributors of YOLOv8, ByteTrack, TensorFlow-Keras, Scikit-Learn, and Firebase, whose tools and documentation were essential to our project's success.

## Abstract

This report presents the development and current status of an AI-based cattle monitoring system aimed at improving livestock management in winter barns. Carried out as part of our bachelor's degree at Høyskolen Kristiania, and in collaboration with Agroidtel: a young and innovative Norwegian startup, this project makes use of advanced artificial intelligence techniques, including the YOLOv8 object detection model and the ByteTrack tracking algorithm, to monitor cattle behavior and identify potential health conditions like sickness or estrus. By combining real-time data collection, thorough analysis, and user-friendly visualization tools, our system is designed to give dairy farmers practical, data-driven insights that support better animal welfare and more efficient operations. The report covers our approach in detail, including our methods, system design, technical choices, evaluation, and ideas for further development, placing our work in the context of ongoing research in AI-powered livestock monitoring. Although we've made significant progress, especially in detection and tracking, the health prediction part of the system still needs more work to reach its full potential. **In the end, we built a working end to end pipeline that showed strong performance in detecting and tracking cattle, analyzing behavior, and laying the groundwork for health predictions, setting the stage for future expansion into full-scale, multi-camera setups in multiple farm environments.**



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# Chapter 1

## Introduction

### 1.1 Background

In recent decades, agriculture has experienced a major shift thanks to the rise of automation, digitalization, and artificial intelligence (AI). These technologies have become key tools for tackling long-standing issues, especially in livestock management, where maintaining both animal welfare and efficient operations is crucial. In the context of dairy farming, keeping an eye on cattle health and behavior inside winter barns is still a time-consuming task. Traditional methods, which mostly rely on manual observation, often fall short - especially with large herds. They're prone to human error and don't provide the kind of real-time feedback that modern farms increasingly rely on.

This gap has sparked growing interest in automated systems that can offer reliable, data-driven insights. In fact, several AI-powered systems are already being used to monitor barn environments, helping ensure consistent data flow under varying conditions [1]. Even governmental organizations are beginning to support AI-based livestock monitoring as a forward-looking solution for large-scale farms [2].

Research also highlights the importance of monitoring systems that combine both environmental and behavioral data to get a full picture of cattle welfare [3, 4]. AI and machine learning (ML) are particularly promising in this space, as they can pick up on subtle behavioral changes that may point to illness or signs of reproduction. Our project builds on this body of work, aiming to use modern AI tools to improve how cattle are monitored in Norwegian winter barns, where animals spend extended periods indoors due to the harsh climate.

### 1.2 Problem Statement

At the heart of our project is a practical and academically relevant question: *"How can we utilize AI and object detection to monitor cattle behavior and health in winter barns to predict whether a cow is sick or in heat?"* This question touches on a major need in today's dairy industry; catching signs of illness or reproductive states early, and it fits well within the scope of a bachelor-level research project. Our goal is to bridge the gap between conventional farming methods and newer, tech-driven solutions, using real-time behavior analysis to build something scalable and suited to the unique environment of winter barns.

## 1.3 Goals

Our work is centered around three key objectives. These were carefully chosen to make sure we add real value for our client, Agroidtel, while also meeting the academic standards of our program at Høyskolen Kristiania:

1. **Develop a Robust Object Detection Model:** Build an AI system that can reliably detect and classify cattle in a barn setting, even under different lighting and crowding conditions.
2. **Implement a Tracking System:** Design a system that can follow the movement of individual cows over time and keep track of their identities accurately.
3. **Analyze Cattle Activity Data:** Collect behavioral data and analyze it in a way that helps predict whether cows are sick, healthy, or in heat, based on their activity patterns.

Together, these goals aim to produce a working prototype that helps optimize how farms are managed, while also letting us apply what we've learned throughout our studies.

## 1.4 Limitations

While the potential of this system is exciting, there are a few important limitations we've had to deal with. Environmental challenges like uneven lighting, animals clustering together, and general camera quality can all reduce detection accuracy. However, our biggest limitation is our current reliance on a single camera feed. Although five Reolink cameras have been installed, we can access only one feed at a time using a screen recording of the Reolink app. This means we are only seeing about a quarter of the barn floor, so cows regularly leave and re-enter the frame, which disrupts continuous tracking and ID consistency.

This main limitation could be solved by locating our system on the same local network as the cameras, inside the barn, or by logging into the farmers' internet router and opening a specific firewall port. Neither of which was desirable to our client. They wanted fully remote access to the camera feeds, which we soon found a method of achieving through Reolinks' tool "Reolink Open Platform" (ROP). However, setting up remote camera feed access with ROP is still a work in progress. Therefore, we are still limited to one camera feed only.

We tried to mitigate this by introducing a re-identification model, but the restricted camera view made it hard to collect enough quality data, which in turn affected how well individual cows could be recognized. Because of this, we also had to create synthetic activity data to train our prediction model, which naturally limits the reliability of our results.

## 1.5 Presentation of the Client/Project

This project was done in partnership with Agroidtel, a growing Norwegian startup founded by Kristian, a 21-year-old entrepreneur with a farming background. Agroidtel's mission is to use AI and object detection to improve livestock management, with a focus on collecting, storing, and analyzing activity data from cattle in winter barns. Our work



aligns closely with their goals, especially when it comes to predicting cattle health and providing farmers with data that they can actually use.

The Agroidtel team currently includes Kristian, along with support from our bachelor group and another team from Høyskolen Kristiania in Bergen, working on an application that will serve as a dashboard for our outputs among many other things. This setup has created a productive and collaborative environment that has added real value to the project.

## 1.6 Relation to Existing Research

Our project is strongly connected to ongoing research in AI for livestock management. Several studies have shown that YOLO(You Only Look Once)-based object detection models work well for identifying and monitoring cattle in real time, even under the less-than-ideal conditions typical of barn environments [5, 6]. There’s also growing evidence that using multiple cameras can significantly improve coverage and tracking accuracy, especially in large or crowded barns [7, 8]. However, managing multiple video streams brings its own technical challenges, which is something we aim to explore in future versions of our system.

Additionally, computer vision approaches to re-identification; matching animals across different frames or camera views, have been studied extensively. These models show promise but are still difficult to apply in practice due to the large data requirements and variability in lighting, posture, and appearance [9, 10, 11]. By grounding our system in this existing research, we hope to provide useful insights into how these techniques can be applied more effectively in everyday farming conditions.

# Chapter 2

## Literature Review

### 2.1 AI in Livestock Management

The application of artificial intelligence (AI) in livestock management has increased significantly in recent years, driven by advances in computational power and sensor technology. AI systems, particularly those utilizing computer vision and machine learning algorithms, have shown substantial promise in automating animal monitoring, behavior recognition, and health prediction tasks. Distante et al. [3] highlight how precision livestock farming integrates these technologies to deliver actionable insights, reduce manual labor, and enhance animal welfare.

Guarnido-Lopez et al. [4] provide an extensive review of computer vision methods for livestock management, indicating the diverse applications from individual animal identification to complex behavior analysis. These technologies enable automated detection of health and welfare indicators such as feeding behavior, social interactions, and movement patterns, significantly outperforming manual observation methods in both accuracy and consistency.

### 2.2 Object Detection Models

Object detection, a fundamental aspect of computer vision, involves identifying and classifying objects within an image or video feed. Among various object detection architectures, YOLO (You Only Look Once) has emerged as a particularly effective model for real-time agricultural applications due to its balance of accuracy and computational efficiency.

Recent studies have validated the effectiveness of YOLO models in cattle detection. Hao et al. [5] demonstrate the performance of YOLOv5-EMA in detecting cattle under complex environmental conditions typical in farms, achieving high mean Average Precision (mAP). Similarly, Arıkan et al. [12] highlight YOLO's capability to detect real-time cattle behaviors such as feeding and resting, critical for automated health monitoring and management decisions. Recent advances in YOLO-based livestock identification also emphasize the importance of tuning for environment-specific variables, such as barn lighting and cow density. [13]

## 2.3 Tracking Algorithms

Tracking algorithms are essential in livestock monitoring to maintain consistent identification of individual animals over time. The performance of tracking algorithms significantly influences the reliability of behavioral data and subsequent predictive analytics.

Traditional tracking algorithms such as SORT (Simple Online and Realtime Tracking) have provided foundational capabilities, but newer algorithms like ByteTrack have introduced improvements in managing occlusions and temporary losses of object detection. The ByteTrack algorithm, in particular, has proven effective in agricultural settings due to its robustness in handling complex animal movements and interactions within densely populated scenes [8, 7].

## 2.4 Behavioral Analysis and Prediction

Behavioral analysis in livestock management involves interpreting patterns in animal behavior to infer health status, predict illnesses, and determine reproductive states such as estrus. AI-driven behavioral analysis systems integrate object detection, tracking, and time-series analytics to provide detailed insights into individual and herd-level behaviors.

Machine learning techniques, including supervised and unsupervised learning algorithms, have been successfully applied to behavioral data. Models trained on activity patterns can accurately predict significant health events, such as lameness or mastitis, days before visible symptoms appear, allowing early interventions [14]. A comprehensive review of smart cattle health monitoring systems underlines the growing role of AI in predicting illness before clinical symptoms appear [15]. Moreover, systems that incorporate multiple sensor modalities (e.g., visual data combined with environmental sensors) significantly enhance predictive capabilities, creating comprehensive monitoring solutions that outperform single-sensor systems [4].

## 2.5 Multi-Camera Systems and Their Challenges

Multi-camera setups in livestock monitoring address limitations inherent to single-camera systems, such as limited field-of-view and occlusions. However, they introduce their own set of complexities, including synchronization, calibration, data integration, and identity preservation across multiple camera feeds.

Han et al. [7] demonstrated the efficacy of multi-camera setups for extensive livestock areas, achieving high tracking accuracy through camera synchronization and advanced data fusion techniques. Despite these advantages, the complexity of managing and processing data from multiple cameras remains challenging. Issues related to camera calibration, consistent lighting conditions, and computational demands must be carefully managed to maintain accuracy and efficiency.

## 2.6 Animal Re-Identification (Re-ID)

Re-identification involves recognizing the same animal across different frames or camera views, an essential capability for continuous behavioral monitoring and accurate individual tracking. Recent research has leveraged deep learning for re-identification, utilizing distinctive visual features to differentiate individuals.

Chen et al. [9] and Meng et al. [10] reviewed several visual re-identification techniques applicable to livestock management, emphasizing the challenges associated with variable animal appearances and environmental factors. Prashant and Sudarshan [11] further discuss advancements in deep learning for animal re-identification, highlighting issues like the need for extensive labeled datasets, susceptibility to occlusions, and variability in animal postures. Despite these challenges, advancements in neural architectures and training methods continue to improve re-ID accuracy, showing promising potential for practical farm deployment.

## 2.7 Summary

This literature review illustrates that AI-driven livestock management systems hold significant promise for enhancing animal welfare and operational efficiency. Object detection, tracking algorithms, behavioral analytics, multi-camera setups, and re-identification methods collectively represent a comprehensive toolkit for addressing challenges in modern dairy farming. Our research aims to integrate and extend these technologies within a practical cattle monitoring system, addressing existing gaps and contributing valuable insights for future advancements.

# Chapter 3

## Methodology

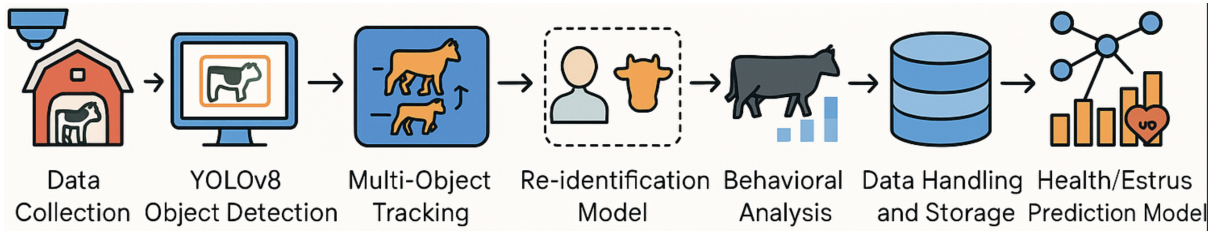


Figure 3.1: Overview of our methodology. The system begins with data collection via barn cameras and proceeds through object detection (YOLOv8), multi-object tracking, re-identification, behavioral analysis, and data handling, culminating in health and estrus prediction using a Conv1D neural network.

### 3.1 Development Approach and Tools

Our development adopted an iterative methodology, emphasizing parallel development and frequent testing to refine each component continuously. We combined agile software practices with experimental evaluations to progressively enhance system robustness and functionality.

Key components of our technical setup included:

- **Object Detection (YOLOv8):** We selected YOLOv8 for cattle detection, leveraging the Ultralytics YOLOv8 framework. This model was chosen due to its balance between high detection accuracy and real-time inference capabilities essential for live cattle monitoring.
- **Multi-Object Tracking (ByteTrack):** To maintain cattle identities consistently across video frames, we implemented ByteTrack. Its robust Kalman-filter-based tracking algorithm effectively handles dynamic and crowded environments.
- **Re-identification (ResNet18-based Classifier):** We experimented with a lightweight Re-ID classifier built on top of a pre-trained ResNet18 model. This allowed us to explore appearance-based identity re-association, with the potential for extending tracking beyond occlusion and re-entry events.
- **Data Handling and Storage (Firebase):** Positional data, activity metrics and health alerts were stored using Firebase Cloud Storage and Firestore, enabling scalable, real-time data retrieval for analytical processes and client interfaces. Most

importantly, the bachelor group in Bergen prefers Firebase as the backend for displaying our outputs in their application.

- **Development Environment:** Python and OpenCV were employed extensively for video processing tasks, supplemented by PyTorch, TensorFlow/Keras, and Jupyter notebooks for machine learning model training. Git was utilized for version control, ensuring efficient collaboration.

## 3.2 Data Annotation and Augmentation

We annotated the initial detection model training images using Roboflow, creating bounding-box annotations for cattle and humans. Although every image had to be manually adjusted and verified, Roboflow was chosen for its easy-to-use tools for auto-annotation, manual annotation and augmentation techniques, which were subsequently applied to enhance data variability.

## 3.3 Real-Time Detection and Tracking Implementation

The live detection and tracking system was built to run continuously on a GPU-enabled Google Cloud VM. The following tools and techniques were used in the pipeline, each selected to overcome specific challenges in real-time cattle monitoring:

- **Google Cloud VM for Deployment:** Google Cloud was used due to free usage credits through Agrolintel and seamless integration with Firebase. The GPU-enabled virtual machine (VM) provided the performance needed for continuous inference and tracking.
- **MSS for Screen Capture:** Due to access restrictions on the RTSP stream, `mss` was used to capture video frames directly from the Reolink desktop application. `pygetwindow` ensured the correct window was targeted automatically.
- **YOLOv8 for Object Detection:** YOLOv8 was chosen for its speed and accuracy in detecting cows and humans. It was run on GPU using the Ultralytics Python package and provided bounding boxes for each frame.
- **ByteTrack for Multi-Object Tracking:** ByteTrack maintained consistent object IDs across frames without relying on appearance-based features. This made it highly suitable for barn conditions with similar-looking animals and frequent occlusions.
- **OpenCV for Zone Classification:** Barn zones (eating, sleeping, null) were defined as polygons, and OpenCV's `pointPolygonTest` was used to classify each cow's location in real time based on their tracked coordinates.
- **Firebase for Data Storage:** Firestore was used to upload per-frame tracking data, including position, zone, and ID. Firebase Storage handled periodic image snapshots for visualization and debugging.

- **System Stability:** Memory management was handled using Python’s `gc` module and `torch.cuda.empty_cache()` to clear GPU memory. `psutil` logged resource usage over time to support performance monitoring.
- **Automated Dataset Generation:** While experimenting with re-identification, cropped cow images were saved periodically into ID-named folders, creating a live-growing dataset for potential future use in re-identification model training.

This toolset allowed the system to function as a scalable and automated real-time pipeline, translating raw video input into structured coordinate and time-in-zone data ready for feature engineering.

### 3.4 Re-identification Model Experimentation

In addition to motion based tracking with ByteTrack, we explored the use of appearance-based re-identification (Re-ID) to improve long-term identity consistency. For this purpose, we used a lightweight image classification model built on **ResNet18**, pre-trained on ImageNet, and fine-tuned on our custom cattle dataset. This approach allowed us to experiment with incorporating visual features, such as coat color and pattern, as a way to recover identity after re-entry or occlusion.

The dataset was organized using a folder-per-identity structure, a format compatible with PyTorch’s `ImageFolder` class. We used `torchvision.transforms` for preprocessing, which included resizing images to 224x224 pixels and normalization. The model was trained using **cross-entropy loss** and the **Adam** optimizer for 20 epochs.

This toolchain was selected for its simplicity, reproducibility, and compatibility with PyTorch workflows already in use for detection and analysis. Although results were preliminary, this exploration provided insight into the potential and limitations of appearance-based Re-ID in barn conditions with limited camera coverage. Details of the training process are provided in `Train_reID.py`.

### 3.5 Behavioral Data Analysis and Feature Engineering

Our behavioral data analysis was implemented through a custom analytical pipeline developed in Python, executed via Jupyter Notebooks. The primary script used for this process was `AgroIntel_Feature_Extraction_Analysis.ipynb`, which orchestrated data retrieval, cleaning, transformation, and export of processed analytics back to the cloud.

We chose Python as the core language for its large ecosystem of scientific libraries, which made it well-suited for numerical analysis and time-series processing. Within Python, the key libraries used were:

- **Pandas:** Provided high-level data structures and tools for reshaping, aggregating, and filtering tabular data. Its `DataFrame` API was crucial for organizing behavioral logs per cow and per time window.

- **NumPy:** Used extensively for numerical operations and efficient array manipulations. It played a foundational role in calculating distances, movement paths, and other geometry-based metrics.
- **SciPy:** Supplemented our analysis with signal processing utilities. We used SciPy’s smoothing filters and interpolation functions to reduce noise and outliers from raw tracking data.
- **Scikit-learn:** Employed for preprocessing tasks such as feature normalization, standardization, and dimensional transformations in preparation for downstream machine learning. Its utilities for scaling and encoding were used before feeding data into the prediction model.
- **Matplotlib and Seaborn:** These libraries were used to visualize behavioral trends and to debug anomalies through graphical inspection.

For cloud integration, the processed features were written back to our Firebase Firestore database, while generated graphs (e.g., movement over time or feature distributions) were uploaded to Firebase Storage.

Our design goal was to create a pipeline that was modular, testable, and generalizable to future data formats. To that end, all intermediate results (e.g., coordinate logs, smoothed positions, time-window splits) were cached using JSON and CSV formats, enabling both reproducibility and auditability.

In summary, the analysis system combined Python’s robust data science stack with Firebase’s real-time cloud infrastructure. This toolchain allowed us to efficiently transform raw coordinate logs into structured activity snapshots for predictive modeling, while preserving flexibility for iterative development.

## 3.6 Predictive Modeling

Our predictive modeling component was developed using Python and TensorFlow, specifically leveraging the Keras Functional API to build a 1D Convolutional Neural Network (Conv1D). This architecture was selected due to its efficiency and suitability for analyzing time-series data, where behavior features evolve sequentially over time. The goal was to classify cattle behavior sequences as indicative of either sickness, estrus, or a healthy state.

Key tools and technologies used include:

- **TensorFlow and Keras:** Provided a flexible deep learning framework for constructing and training the neural network. The Keras Functional API allowed for a clear definition of complex model architectures, including convolutional layers, batch normalization, global average pooling, and multiple output heads.
- **NumPy and Pandas:** Used for preparing the input time-series data. These libraries helped reshape behavioral features into 3D tensors of shape (samples, time steps, features), required for Conv1D input. They were also used to load and manipulate CSV-based training data generated by the analysis pipeline.
- **Scikit-learn:** Employed for preprocessing tasks, including scaling of features to standardized ranges and shuffling of training data. Additionally, we used it for splitting the dataset into training and validation sets.



- **Matplotlib:** Used to visualize training progress over epochs, including accuracy and loss trends, which were critical in diagnosing overfitting and underfitting during experimentation.

Due to the lack of real activity data and health labels, we simulated training targets using heuristic rules applied to behavioral patterns: low movement and prolonged inactivity were labeled as "sick", while high activity and frequent transitions were labeled as "estrus". This synthetic labeling allowed us to prototype and evaluate the model pipeline without requiring clinical annotations.

All model artifacts, including training histories and weight files, were saved locally and can be uploaded to Firebase Storage for inference use in the full system loop. The modularity of the model design allows for replacement with more complex architectures once real labeled data becomes available.

Overall, the toolchain enabled fast iteration, explainable modeling, and integration-ready outputs, making it a practical foundation for future expansion into production-level health prediction systems.

### 3.7 Justification and Alternative Choices

All tools and methods were selected through a balance of empirical testing, system requirements, and compatibility with the real-world constraints of the barn environment. Below, we summarize the rationale for each major methodological choice, alongside evaluation of relevant alternatives.

- **YOLOv8 for Object Detection:** We selected YOLOv8 for its strong trade-off between speed, accuracy, and deployment simplicity. Alternatives such as Faster R-CNN achieved higher accuracy in static benchmarks but were computationally expensive and unsuitable for real-time inference. SSD offered better speed but lacked the precision needed to detect cows in crowded or occluded conditions. YOLOv8's architecture and pre-trained weights provided excellent out-of-the-box performance, making it ideal for real-time barn monitoring with minimal tuning.
- **ByteTrack for Multi-Object Tracking:** ByteTrack was chosen for its robustness in cluttered environments with many similar-looking objects. It maintained stable identities through Kalman filtering and a matching mechanism that does not rely on appearance features. We initially tested DeepSORT, which includes appearance embeddings, but it performed poorly under barn lighting and occlusion. ByteTrack required less compute and produced more consistent identity tracking across sequences, making it more practical for our prototype phase.
- **Conv1D for Predictive Modeling:** The Conv1D architecture was selected due to its ability to efficiently learn local temporal patterns in behavioral time series data. Unlike RNN-based approaches (e.g., LSTM or GRU), Conv1D networks are faster to train, easier to interpret, and better suited for short fixed-length sequences like the 6-hour activity windows we used. Transformer models were also considered, but their complexity and data requirements exceeded the scope of our current dataset. Conv1D provided an optimal starting point with clear future extension paths.

- **Firestore for Data Handling and Integration:** Firestore was chosen as a unified platform for data storage, synchronization, and client access. We considered traditional SQL databases (e.g., PostgreSQL) for structured logging, but they lacked native support for real-time sync and required additional server-side infrastructure. Firestore allowed direct communication with cloud functions and front-end clients, including the mobile dashboard being developed by our partner group in Bergen. The platform’s scalability and zero-maintenance deployment model significantly accelerated system development.
- **Google Cloud Platform (GCP) for Deployment:** We ran our detection and tracking pipeline on a GPU-enabled VM hosted on GCP. This decision was influenced by Agrosoft’s existing partnership with Google, which provided us with free cloud credits. GCP also offered close integration with Firestore, allowing our system to transfer data between components efficiently without complex authentication or networking setups. Other cloud platforms like AWS and Azure were viable but would have introduced more configuration overhead.
- **Roboflow for Annotation and Augmentation:** Roboflow was used for labeling and augmenting training images. Compared to alternatives like CVAT or LabelImg, Roboflow provided a faster workflow and built-in augmentation features, including brightness, blur, and hue shift operations. Its ability to export directly in YOLO format reduced preprocessing steps, and its dataset versioning system made it easier to iterate and retrain models quickly.
- **OpenCV for Geometric Zone Mapping:** Zone classification (eating, sleeping, null) was done using polygon-based mapping with OpenCV’s `pointPolygonTest`. Alternatives like image segmentation or deep learning-based zone classifiers were deemed too complex for the pixel-accurate and static zone layout in our setup. The OpenCV method was efficient, deterministic, and required no additional training.

These decisions reflect a strategic balance between cutting-edge research practices and the constraints of a real-world barn environment. By combining proven tools with pragmatic engineering, we ensured that the system remained robust, interpretable, and easily extendable for future iterations.

### 3.8 Methodological Critique and Project Limitations

While robust, our approach presented several methodological and practical constraints:

- Dependency on single-camera data significantly restricted comprehensive cattle monitoring and re-identification capabilities.
- Real-time processing relied on video screen recording instead of direct RTSP stream access, introducing potential inefficiencies and limitations.
- Dataset and model validation were limited to one barn environment, raising concerns about generalizability.
- The predictive model, currently trained on synthetically derived datasets, requires further validation against real cattle health events to confirm its practical applicability.

These limitations outline clear directions for future enhancements, particularly concerning multi-camera integration, direct stream utilization, and validation with expanded real-world data.

## 3.9 System Analysis and Design

### 3.9.1 Requirements Analysis

We identified key system requirements to achieve our project goals effectively:

- Real-time detection and differentiation of cows and humans from barn camera feeds.
- Robust and consistent tracking of individual cows over time.
- Efficient and scalable storage for positional and behavioral data.
- Automated extraction of actionable behavioral insights from raw data.
- Reliable prediction of health and estrus states using extracted behavioral features.
- User-friendly presentation of insights and alerts for farmers.

### 3.9.2 System Architecture

Our system follows a modular, scalable architecture comprising multiple interconnected components:

- **Cameras (Barn Overhead):** We use overhead-mounted Reolink cameras, providing wide-area coverage of barn activity. Video streams make up the initial input for all further processing.
- **Detection (YOLOv8):** Video frames captured from cameras are processed using the YOLOv8 model, which accurately detects cows and humans in real-time and draws bounding-boxes around each class.
- **Tracking (ByteTrack):** Detection outputs feed into the ByteTrack algorithm, which assigns and maintains consistent identities for each cow. ByteTrack is also used to locate the centre of each bounding box for coordinate collection and zone determination.
- **Realtime Database (Firestore):** Tracked positional data (coordinates, timestamps, zone presence), the activity data derived thereof and the prediction alerts are stored in Firestore, a cloud-based real-time database.
- **Analysis & Feature Engineering:** The analysis engine retrieves raw coordinate data from Firestore and computes detailed behavioral metrics, such as total distance moved, speed, zone occupancy duration, and behavioral transitions. Features are engineered hourly and cumulatively and uploaded to firestore to prepare data for prediction tasks.

- **Health/Estrus Prediction Model (Conv1D):** Extracted behavioral features serve as inputs for our Conv1D neural network prediction model. This model identifies patterns indicative of health conditions (e.g., illness or estrus) based on temporal behavioral changes.
- **Farmer Dashboard (Alerts & Monitoring):** Predictions and relevant behavioral insights are visualized through a user-friendly dashboard. Farmers receive real-time alerts about significant health and behavioral events, facilitating timely and informed interventions. The dashboard is being developed by another bachelor group at Høyskolen Kristiania in Bergen, and our system has been designed to integrate seamlessly with their application.

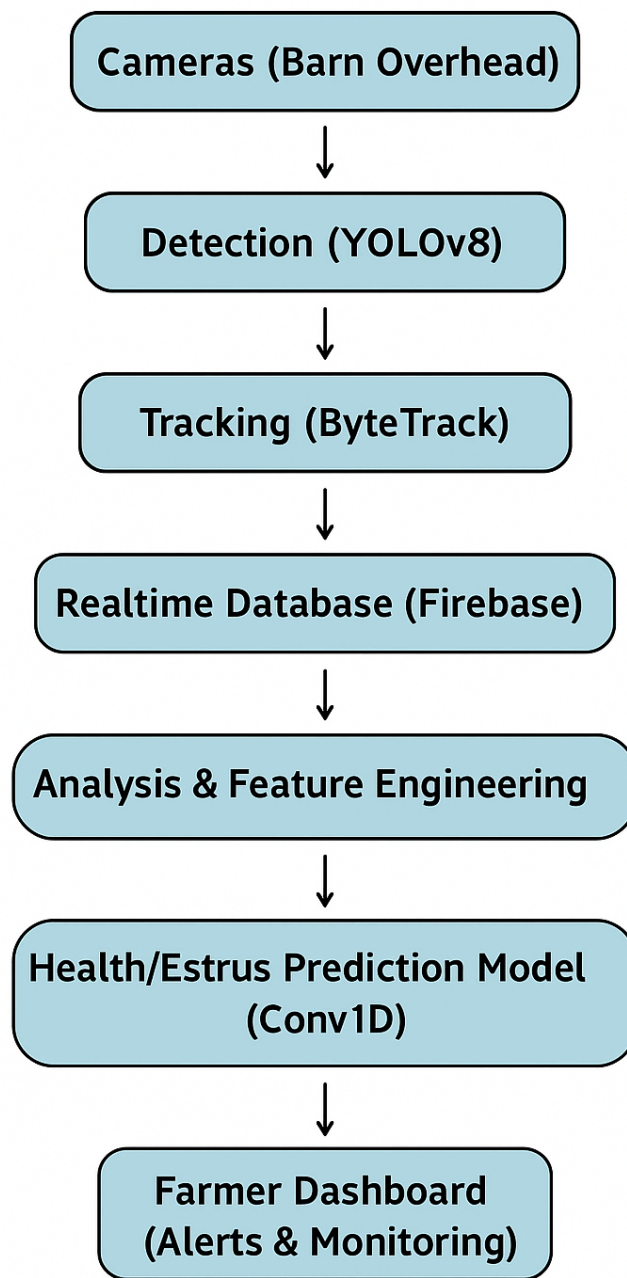


Figure 3.2: System Architecture Overview

# Chapter 4

## Implementation

### 4.1 Camera Installation

The first step of our project was to mount cameras in the roof of the farmers barn. We mounted five Reolink cameras to the ceiling using a stable metal scaffold placed over the central alleyway. The elevated position provided an unobstructed top-down view of the feeding and resting areas, minimizing occlusion and maximizing detection accuracy. Although five cameras were installed, we only have access to one of the video streams at a time for now.



Figure 4.1: Installing the overhead Reolink cameras in the barn.





Figure 4.2: Footage from the roof mounted camera.

## 4.2 Dataset Preparation and Annotation

We collected video footage from a single overhead-mounted Reolink camera placed in the barn. From this feed, we extracted still frames and initially used Roboflows’ auto-annotation tool to annotate them, before manually adjusting and verifying every bounding box. Bounding boxes were applied to cows and humans, creating a labeled dataset for object detection training. While the dataset was limited in scale, we ensured that it captured varied lighting, occlusion, and pose conditions.

To increase model generalization, we applied several augmentation techniques via Roboflow:

- Brightness and contrast adjustment to simulate different lighting environments
- Gaussian blur to simulate focus issues
- Noise to ensure robustness

The dataset, including annotations, was exported in YOLO format and prepared for training with Ultralytics’ YOLOv8 framework.

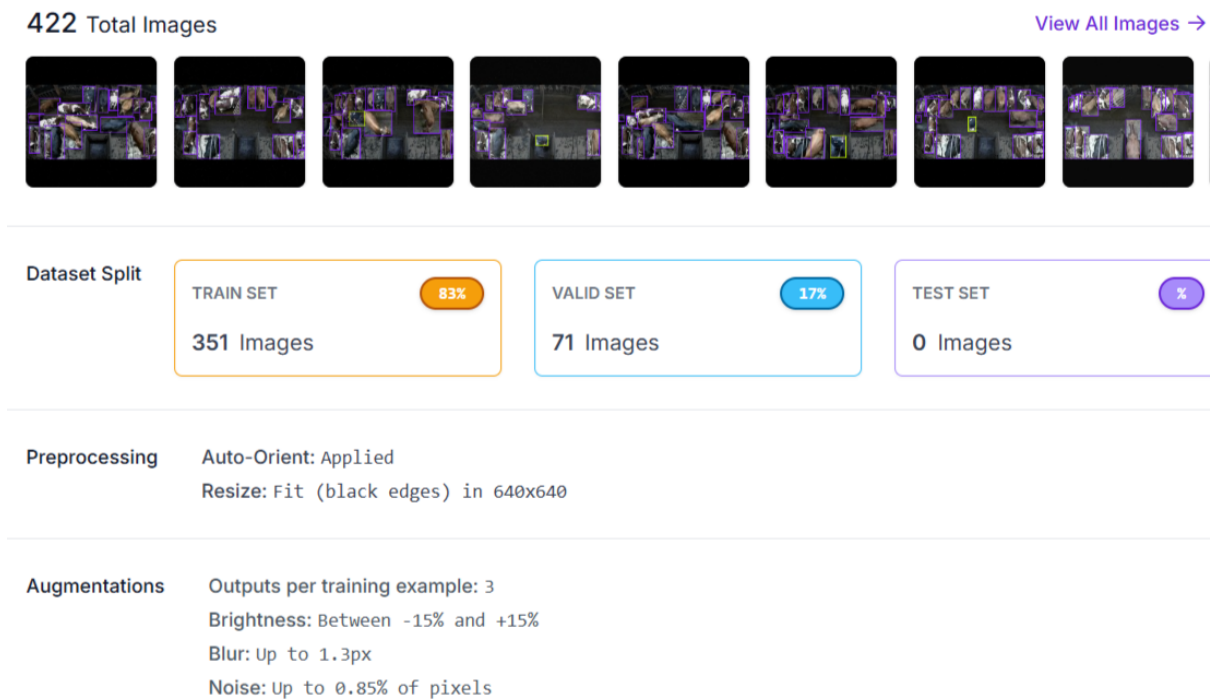


Figure 4.3: Roboflow dataset overview. The dataset consists of 422 total images, with 351 (83%) used for training and 71 (17%) for validation.

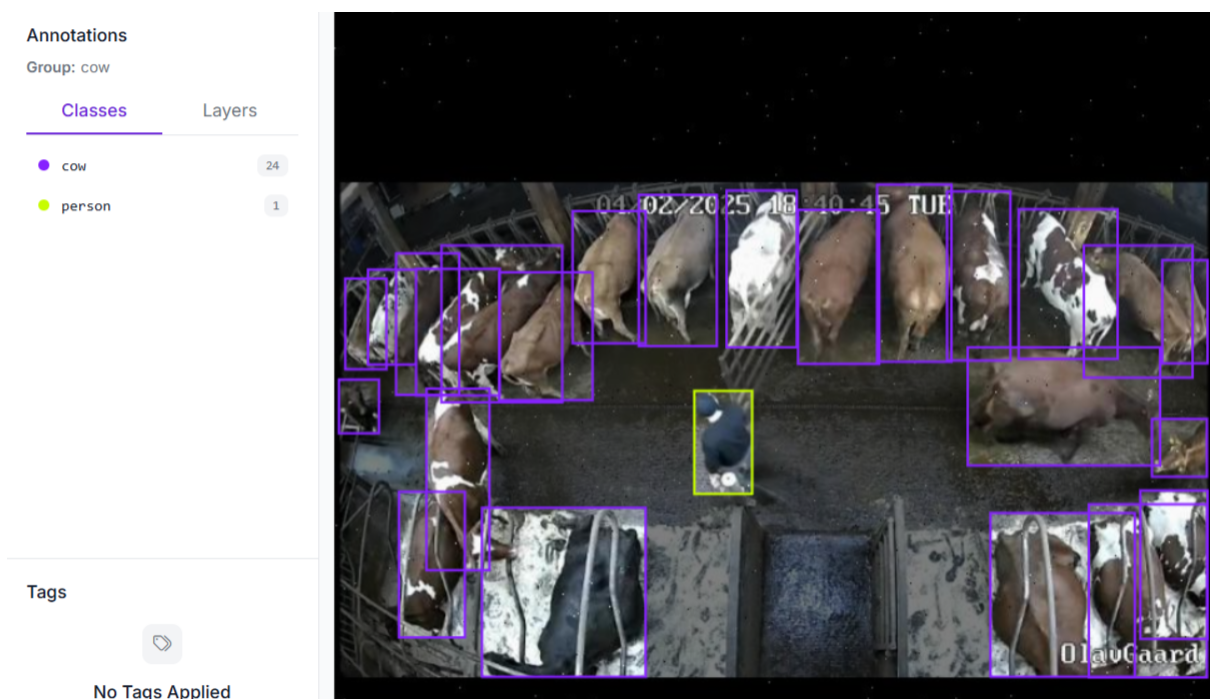


Figure 4.4: Annotated image from the Roboflow labeling interface. Bounding boxes are applied to 24 cows and one person.



## 4.3 Object Detection Model Training

We used the script `AgroIntel_YOLOv8TrainModel.ipynb` to train our cattle detection model. The training was conducted using the YOLOv8 architecture from the Ultralytics library, utilizing GPU acceleration on Google Colab for efficiency.

The model was designed to detect two classes: cows and humans. Before starting the training, we set key hyperparameters such as image resolution (640 pixels), batch size, learning rate, and data augmentation settings. Augmentation techniques like brightness adjustments, Gaussian noise, and random blurring were applied to make the model more robust against varying lighting and visual conditions found in barns.

We trained the model for 50 epochs, closely monitoring performance metrics such as loss, precision, recall, and mean Average Precision (mAP@0.5). Early epochs showed rapid improvement, with performance stabilizing towards the end of training. We also tuned anchor boxes-predefined bounding box shapes-to better match the size and shape of cows and humans in our data.

After training, we evaluated the final model performance on the validation set to confirm high accuracy and low false positives. The best-performing weights were then exported as a `.pt` file, ready for use in real-time cattle monitoring. This model served as the core detector for all real-time video processing and tracking tasks

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
49/50   2.49G    0.6237   0.4829   0.9405    459        640: 100%|██████████| 22/22 [00:01<00:00, 12.16it/s]
      Class  Images  Instances  Box(P)    R    mAP50  mAP50-95): 100%|██████████| 3/3 [00:00<00:00, 5.04it/s]
      all      71      1582      0.98    0.95    0.983    0.714

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
50/50   2.47G    0.6226   0.4835   0.9429    446        640: 100%|██████████| 22/22 [00:01<00:00, 12.32it/s]
      Class  Images  Instances  Box(P)    R    mAP50  mAP50-95): 100%|██████████| 3/3 [00:01<00:00, 1.84it/s]
      all      71      1582      0.978    0.951    0.982    0.714

50 epochs completed in 0.041 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 6.2MB
Optimizer stripped from runs/detect/train/weights/best.pt, 6.2MB

Validating runs/detect/train/weights/best.pt...
Ultralytics YOLOv8.0.81 Python-3.11.11 torch-2.0.1+cu118 CUDA:0 (NVIDIA A100-SXM4-40GB, 40507MiB)
Model summary (fused): 168 layers, 3006038 parameters, 0 gradients, 8.1 GFLOPs
      Class  Images  Instances  Box(P)    R    mAP50  mAP50-95): 100%|██████████| 3/3 [00:01<00:00, 1.84it/s]
      all      71      1582      0.961    0.955    0.986    0.731
      cow      71      1541      0.948    0.959    0.98    0.817
      person  71         41      0.974    0.951    0.991    0.644
Speed: 2.0ms preprocess, 0.6ms inference, 0.0ms loss, 1.5ms postprocess per image
Results saved to runs/detect/train

```

Figure 4.5: Training output of the YOLOv8 model.

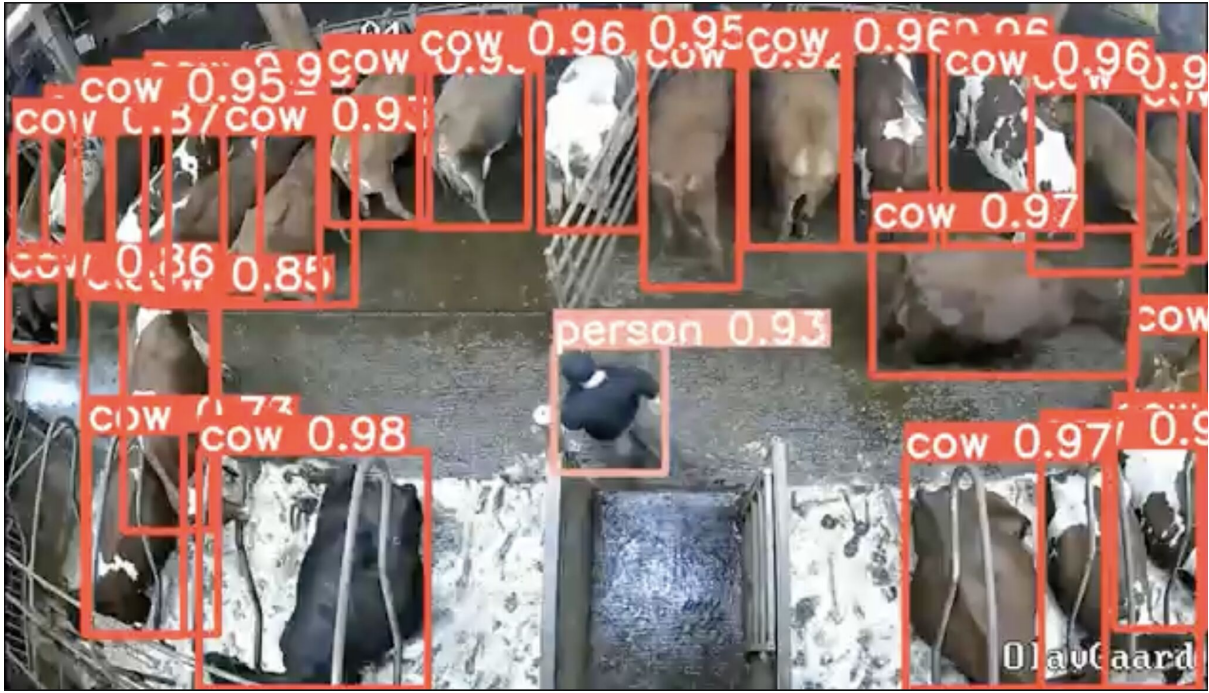


Figure 4.6: Detection output of the YOLOv8 model.

## 4.4 Live Detection and Tracking Pipeline

The real-time pipeline was implemented in the script `Detection_Tracking.py`, running on a GPU-enabled virtual machine hosted on Google Cloud. Due to restrictions accessing the RTSP stream remotely, we used the Reolink desktop application to view the camera feed, and employed the `mss` screen-capture library to continuously grab frames.

Each frame was passed to the trained YOLOv8 model, which returned bounding boxes and class labels. These detections were then processed by the ByteTrack tracker. ByteTrack assigned and maintained unique IDs for each cow across frames, handling occlusion and reappearance effectively.

Each cow's position was mapped onto predefined pixel-based zones:

- Eating zone
- Sleeping zone
- Null zone (Anything outside the other zones)

For each detection, we recorded:

- ID, (x, y) center position
- Assigned zone
- Frame timestamp

This data was continuously written to Firebase Firestore. Snapshot images and videos were also periodically saved and uploaded for debugging or visualization.

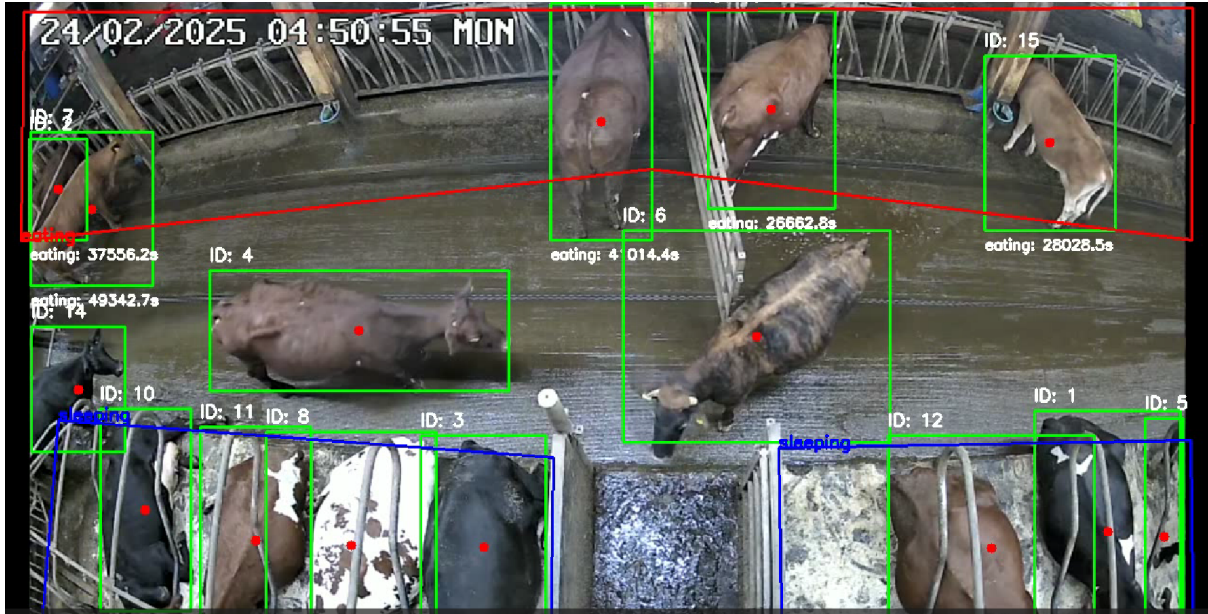


Figure 4.7: Live detection and tracking output showing zone assignments, cow IDs, bounding boxes and time-in-zone statistics. Red is the eating zone, blue is the sleeping zone and everything outside of them is considered "null"-zone.

```

0: 480x640 4 cows, 11.2ms
Speed: 4.1ms preprocess, 11.2ms inference, 1.9ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 4 cows, 11.7ms
Speed: 3.6ms preprocess, 11.7ms inference, 2.0ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 4 cows, 12.1ms
Speed: 3.6ms preprocess, 12.1ms inference, 1.7ms postprocess per image at shape (1, 3, 480, 640)

0: 480x640 5 cows, 12.8ms
Speed: 3.6ms preprocess, 12.8ms inference, 2.0ms postprocess per image at shape (1, 3, 480, 640)
Collecting coords at 2025-04-28T22:35:37Z

```

Figure 4.8: Console output during real-time tracking, displaying detection and tracking speed metrics. Inference maintained an average processing time of approximately 12–13ms per frame, including preprocessing and postprocessing.

|                             |                      |                                   |
|-----------------------------|----------------------|-----------------------------------|
| activity_snapshots          | 1                    | coordinates                       |
| alerts                      | 10                   |                                   |
| cows                        | 11                   | + Add field                       |
|                             | 12                   | time_in_zone                      |
|                             | 13                   | eating: 10479.435762643814        |
|                             | 14                   | sleeping: 3095.882290840149       |
|                             | 15                   |                                   |
| coordinates                 | 2025-04-29T16:57:17Z | + Add field                       |
|                             | 2025-04-29T16:57:19Z | coordinates                       |
| Add field                   | 2025-04-29T16:57:21Z | x: 313                            |
| time_in_zone                | 2025-04-29T16:57:23Z | y: 116                            |
| eating: 10479.435762643814  | 2025-04-29T16:57:26Z | timestamp: "2025-04-29T16:57:17Z" |
| sleeping: 3095.882290840149 | 2025-04-29T16:57:28Z | zone: "Null Zone"                 |
|                             | 2025-04-29T16:57:30Z |                                   |

Figure 4.9: Example of coordinate data stored in Firebase for one cow. The database tracks each cow’s total zone time (e.g., eating, sleeping) and logs detailed time-stamped coordinates alongside the identified zone (e.g., Null Zone).

## 4.5 Re-identification Model Integration

To improve long-term identity assignment, especially when cows leave and re-enter the frame, we integrated an appearance based re-identification (Re-ID) model alongside the existing YOLOv8 + ByteTrack tracking system.

The re-identification model was trained using the script `Train_reID.py`. This script loaded our cattle image dataset, organized into folders per cow ID, and fine-tuned a ResNet18 model pre-trained on ImageNet. The images were transformed using `torchvision.transforms` (resizing, normalization) and fed into a PyTorch `DataLoader` for batched training. We used a softmax classifier head and cross-entropy loss, optimized with Adam over 20 epochs. The trained model weights were saved as `cattle_reid.pth`.

During live deployment, the script `Tracking_reID.py` was used to run the Re-ID model in parallel with YOLOv8 and ByteTrack. After initial detection and tracking, cropped cow images were generated in real time from the bounding boxes of each detected cow. These cropped images were resized and passed through the ResNet model to generate a predicted identity label. If the cow had previously disappeared and was assigned a new ByteTrack ID upon re-entry, the Re-ID model attempted to recover and reassociate the original identity.

To enable this logic, we implemented a lookup mechanism comparing current ByteTrack IDs with re-identified labels. If a visual match was found (i.e., the predicted label matched an earlier one), we reassigned the old ID to the current track. We also logged all predictions, model confidence, and frame metadata for validation and debugging.

However, performance was limited by the constrained field of view of the single camera. Many cows were never visible in full, and the model struggled with visually similar individuals. Additionally, the added inference time from the Re-ID model introduced latency, increasing frame processing time from 13ms to over 2 seconds in some cases.

Although ultimately excluded from the main pipeline due to these performance issues,

the integration serves as a functional proof of concept. Once additional camera feeds are accessible, this module can be revisited and improved.

## 4.6 Behavioral Data Processing and Feature Engineering

The script `AgroIntel_Feature_Extraction_Analysis.ipynb` handles the analysis of cattle behavior after tracking. It fetches the cows' positional coordinates and zone data from Firebase and smooths the data to reduce noise and inaccuracies.

To translate these coordinates into real-world measurements, we calibrated the video frames by measuring a known distance on the barn floor. We found that about 100 pixels represented one meter near the center of the camera view. However, due to lens distortion and perspective differences, the scale was slightly smaller near the edges of the frame (around 85–90 pixels per meter). To fix this, we set multiple calibration points throughout the frame and used spatial interpolation to adjust our measurements depending on where each cow was located.

We extracted the following behavioral features for each cow:

- Total distance moved
- Time spent in each zone
- Frequency of zone transitions
- Average movement speed
- How often and for how long cows stopped moving
- Turning angles and how irregular their paths were
- Resting periods, bursts of activity, and approximate distance to their nearest neighbor

We calculated distances accurately using our calibrated pixel-to-meter conversions. Movement speed and stops were identified by analyzing how much the cow's position changed between frames, applying thresholds to the smoothed data. Turning angles and path irregularity (tortuosity) were calculated by looking at directional changes over time.

These features were computed both hourly and as cumulative totals. The results were uploaded back to Firebase, creating detailed behavioral profiles used for prediction and visualization.

Additionally, we generated graphs and plots from the activity data, storing them in Firebase storage. Examples of these visualizations, which include both herd-wide summaries and individual cow activities, can be seen in Appendix B.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                          |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------|-----------|
| activity_snapshots                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | > | ⋮ 2025-04-29T19:15:25Z > | snapshots |
| alerts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   | 2025-04-29T19:30:31Z     |           |
| cows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   | 2025-04-29T20:35:17Z     |           |
| <pre> cow_id: "1" historical_avg_net_displacement: 2.107293961033066 historical_avg_num_stops: 12.625 historical_avg_resting_hours: 0.15201388888888887 historical_avg_speed: 0.011589287752829507 historical_avg_stop_duration_s: 2.248763615497587 historical_avg_time_eating_hours: 0.1488888888888889 historical_avg_time_null_zone_hours: 0.02826388888888887 historical_avg_time_sleeping_hours: 0.01493055555555556 historical_avg_tortuosity: 4.468222891595875 historical_avg_total_distance_per_hour: 8.16975307715388 historical_avg_turning_angle_variance: 2.328018829735684 historical_avg_zone_transitions: 6.75 historical_hours_count: 4 historical_total_distance: 32.67901230861552 historical_total_num_stops: 50.5 historical_total_resting_hours: 0.6080555555555555 historical_total_time_eating_hours: 0.5955555555555556 historical_total_time_null_zone_hours: 0.11305555555555555 historical_total_time_sleeping_hours: 0.05972222222222225 historical_total_zone_transitions: 27 hourly_avg_nearest_neighbor_distance: 0.011298497300302815 hourly_avg_speed: 0.04753517070502067 hourly_avg_stop_duration_s: 2.359924026590693 hourly_net_displacement: 2.415461032598125 hourly_num_stops: 88.5 hourly_resting_hours: 0.6180555555555556 hourly_time_eating_hours: 0.7363888888888889 hourly_time_null_zone_hours: 0.13583333333333333 hourly_time_sleeping_hours: 0.10333333333333333 hourly_tortuosity: 69.20175804216575 hourly_total_distance: 167.1541499381353 hourly_turning_angle_variance: 2.1813185428838913 hourly_zone_transitions: 63 timestamp: "2025-04-29T19:15:25Z" </pre> |   |                          |           |

Figure 4.10: Snapshot of the Firebase `activity_snapshots` database. Each entry contains both historical and hourly behavioral metrics for individual cows, including movement statistics, zone transitions, and derived features such as tortuosity and stop frequency. These values are used as input to the Conv1D prediction model.

## 4.7 Health and Estrus Prediction Model

We implemented the final prediction stage in `AgroIntel_Prediction_Model.ipynb`, where the system interprets each cow’s behavior to detect potential health issues or estrus (heat) events. Since real-world labeled health data was unavailable, we simulated labels based on behavioral assumptions: low movement combined with extended resting time was used to represent sickness, while high movement with frequent zone transitions was used to represent estrus. These assumptions allowed us to create a synthetic dataset that mimics realistic activity profiles and their corresponding health states based on our activity features. With a multi-camera set up and correct activity data, we will be able to add true data and health labels as the farmer knows which cow was sick/in heat and when.

The model uses a 1D Convolutional Neural Network (Conv1D), built with TensorFlow and Keras, to classify behavior across three categories: *healthy*, *sick*, or *in estrus*. Each input sample represents a 6-hour time window per cow, with 33 distinct behavioral features, including totals, average totals and hourly features. These include displacement, number of stops, time spent eating or sleeping, zone transitions, and more. To make the synthetic data more robust and reflective of real-world variability, we injected label noise and feature-level outliers. The resulting dataset was then standardized, encoded, and split into training, validation, and test sets.

```
x: (15000, 198) labels: (array(['healthy', 'in_heat', 'sick'], dtype='<U7'), array([5003, 4996, 5001]))

Train: (10800, 6, 33) (10800, 3)
Val:   (1200, 6, 33) (1200, 3)
Hold:  (3000, 6, 33) (3000,)
```

Figure 4.11: Overview of the generated synthetic dataset. The distribution shows an even split of 15 000 generated activity snapshots among the three classes: healthy, in\_heat, and sick. The final dataset was divided into training, validation, and holdout sets with shapes (10800, 6, 33), (1200, 6, 33), and (3000, 6, 33) respectively.

The Conv1D architecture was chosen for its ability to detect local patterns in time-series data while remaining computationally efficient and relatively interpretable. It consists of two convolutional layers that scan across the time dimension, followed by batch normalization, global average pooling, and a dense output layer. The model outputs three classification logits (healthy, sick and in heat). We trained using categorical crossentropy with label smoothing, optimized with the Adam optimizer, and monitored using early stopping and learning rate scheduling to ensure stable convergence.

Model: "functional"

| Layer (type)                                        | Output Shape  | Param # |
|-----------------------------------------------------|---------------|---------|
| input_layer_3 (InputLayer)                          | (None, 6, 33) | 0       |
| gaussian_noise_3 (GaussianNoise)                    | (None, 6, 33) | 0       |
| conv1d_6 (Conv1D)                                   | (None, 6, 64) | 6,400   |
| batch_normalization_6 (BatchNormalization)          | (None, 6, 64) | 256     |
| conv1d_7 (Conv1D)                                   | (None, 6, 64) | 12,352  |
| batch_normalization_7 (BatchNormalization)          | (None, 6, 64) | 256     |
| global_average_pooling1d_3 (GlobalAveragePooling1D) | (None, 64)    | 0       |
| dense_3 (Dense)                                     | (None, 128)   | 8,320   |
| dropout_3 (Dropout)                                 | (None, 128)   | 0       |
| dense_4 (Dense)                                     | (None, 3)     | 387     |

**Total params:** 27,971 (109.26 KB)

**Trainable params:** 27,715 (108.26 KB)

**Non-trainable params:** 256 (1.00 KB)

Figure 4.12: Architecture of the Conv1D health and estrus prediction model.

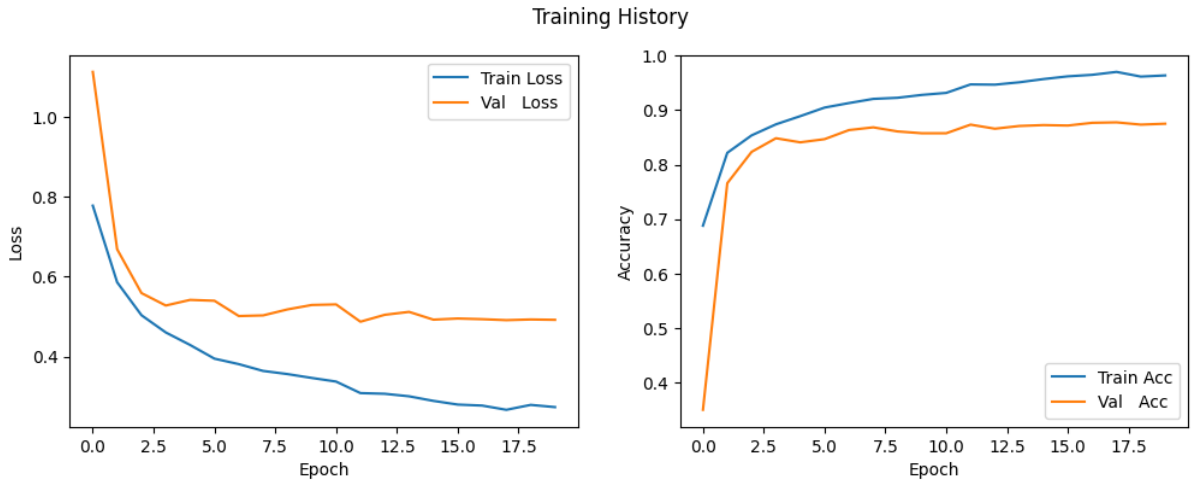


Figure 4.13: Training and validation performance of the Conv1D model over 20 epochs. The loss steadily decreases while accuracy increases, indicating effective learning and convergence. However, the training and validation discrepancy shows signs of overfitting.

During inference, the trained model processes cow behavior data in near real time. For each cow, it outputs a predicted label and associated class probabilities. In cases where the prediction indicates a non-healthy condition, a saliency map is generated to highlight



the most influential features, providing an interpretable explanation of the decision. This not only builds trust among farmers and veterinarians but also increases the system's transparency.

```

=== Batch @ 2025-04-29 12:42:50.064734 ===
Cow 6 | True: in_heat → Pred: in_heat (p=0.99)
Cow 30 | True: healthy → Pred: healthy (p=0.92)
Cow 3 | True: in_heat → Pred: in_heat (p=0.88)
Cow 26 | True: in_heat → Pred: in_heat (p=0.96)
Cow 25 | True: in_heat → Pred: in_heat (p=0.91)
Cow 11 | True: sick → Pred: sick (p=0.97)
Cow 17 | True: in_heat → Pred: in_heat (p=0.97)
Cow 15 | True: in_heat → Pred: in_heat (p=0.90)
Cow 12 | True: healthy → Pred: healthy (p=0.97)
Cow 32 | True: sick → Pred: sick (p=0.98)
Cow 34 | True: healthy → Pred: sick (p=0.95)
Cow 17 | True: healthy → Pred: healthy (p=0.92)
Cow 28 | True: sick → Pred: sick (p=0.88)
Cow 22 | True: healthy → Pred: sick (p=0.98)

⚠ ALERT: Cow 6 → in_heat (p=0.99)
• historical_avg_num_stops: 0.004
• historical_avg_resting_hours: 0.003
• historical_avg_speed: 0.004
• historical_avg_stop_duration_s: 0.003
• historical_avg_time_sleeping_hours: 0.004
• historical_avg_total_distance_per_hour: 0.003
• historical_avg_turning_angle_variance: 0.002
• historical_avg_zone_transitions: 0.003
• historical_total_distance: 0.002
• historical_total_time_sleeping_hours: 0.002
• hourly_avg_stop_duration_s: 0.002
• hourly_num_stops: 0.003
• hourly_resting_hours: 0.002
• hourly_time_eating_hours: 0.003
• hourly_time_sleeping_hours: 0.002
• hourly_tortuosity: 0.006

```

Figure 4.14: Example output from the Conv1D prediction model's inference loop. The left panel shows batch predictions, including confidence scores. The right panel displays the detailed alert and contributing features for the top prediction, indicating Cow 6 is in heat with 99% confidence.

Finally, the prediction results, along with the classification label, confidence score, and key influencing features, are sent to a Firebase database. These are displayed on the farmer dashboard to enable proactive health monitoring and intervention. This predictive capability transforms the system from a passive data logger into a real-time decision support tool, reducing the reliance on manual inspections and enabling more efficient livestock management.

```
confidence: 0.9605751037597656
cow_id: "41"
feature_impacts
  ▼ 0
    feature: "historical_avg_num_stops"
    impact: 0.0031154330354183912
  ▼ 1
    feature: "historical_avg_stop_duration_s"
    impact: 0.004690577741712332
  ▼ 12
    feature: "hourly_time_sleeping_hours"
    impact: 0.006046259310096502
  ▼ 13
    feature: "hourly_turning_angle_variance"
    impact: 0.004530657548457384
predicted_label: "sick"
timestamp: April 29, 2025 at 2:44:10 PM UTC+2
```

Figure 4.15: Example of a prediction alert stored in Firebase. The system predicted the cow labeled as ID 41 to be potentially sick with a confidence of 96%. The contributing features include sleeping time, turning angle variance, and stop frequency. The timestamped alert is stored with accompanying interpretability values.

## 4.8 Firebase Integration

All core components interact with Firebase:

- **Firestore:** Stores live coordinate and zone data per cow
- **Storage:** Used for uploading graphs to visualize the behavioral activity data.

- **Activity Snapshots:** Hourly and total summaries of each cow’s behavior
- **Predictions:** Health and estrus outputs stored for frontend consumption

Firebase provides scalability, near real-time access, and straightforward integration with mobile or web dashboards.

## 4.9 Summary

Our implementation includes data collection, object detection, multi-object tracking, zone-based behavior classification, feature engineering, predictive modeling, and cloud integration. All components are modular and work together in a pipeline.

Despite being limited to a single camera feed and synthetic prediction labels, the system demonstrated the ability to process live video into meaningful behavioral summaries and health-related insights. It successfully tracked cattle across time and space, extracted relevant features, and used these to simulate health state predictions with explainable outputs.

Our approach prioritized real-time performance, cloud scalability, and interpretability. The system is fully operational, and with access to additional camera streams and real-world health annotations, it can scale into a robust, production-ready platform.

# Chapter 5

## Evaluation

We evaluated our cattle monitoring system across technical performance, practical utility, and alignment with both research and client goals. The results show promising outcomes and highlight areas for future development.

### 5.1 Technical Performance Assessment

#### Object Detection and Tracking Performance

The performance of our YOLOv8 object detection model was evaluated on a held-out validation set comprising 71 annotated frames. The model demonstrated high precision and recall across both target classes (cows and humans), with mean Average Precision (mAP@0.5) scores nearing optimal levels. These metrics are summarized in the table below:

Table 5.1: YOLOv8 Detection Performance on Validation Set

| Class   | Precision | Recall | mAP@0.5 |
|---------|-----------|--------|---------|
| Cow     | 0.984     | 0.959  | 0.980   |
| Person  | 0.991     | 0.961  | 0.991   |
| Overall | 0.987     | 0.951  | 0.986   |

The high mAP values confirm the model’s ability to distinguish between cows and humans under variable lighting and occlusion. Additionally, we visualized performance with a confusion matrix (Figure 5.1), which shows minimal cross-class misclassification and near-perfect background rejection:

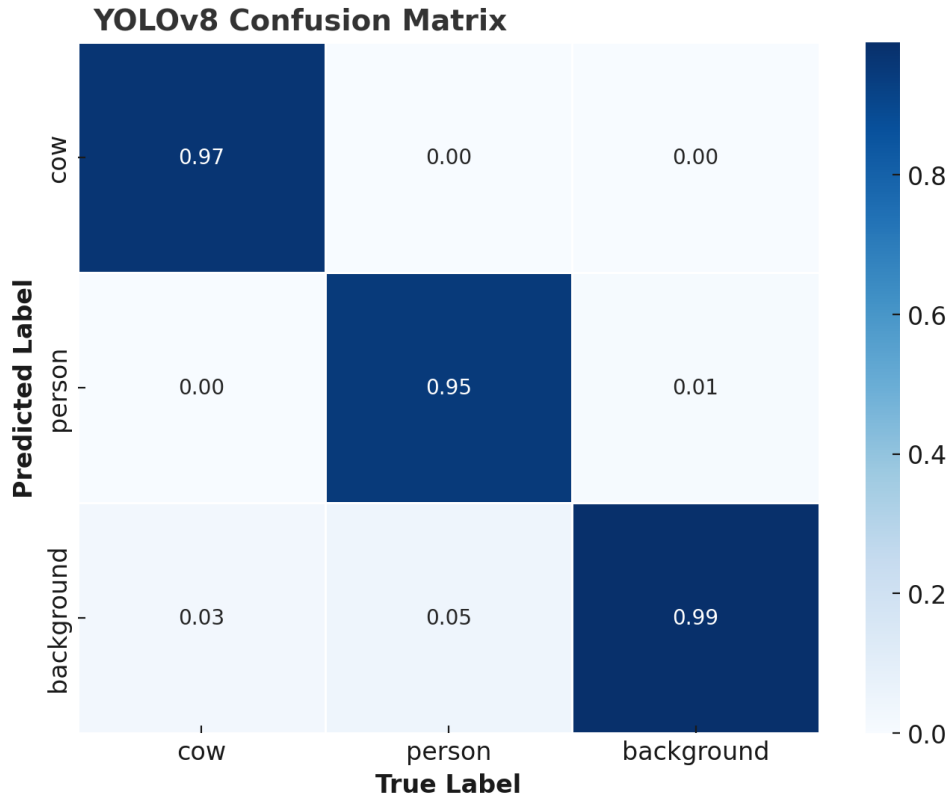


Figure 5.1: YOLOv8 Confusion Matrix on Validation Data

### Tracking Reliability

Tracking evaluation was conducted qualitatively by reviewing hours of recorded footage. ByteTrack maintained consistent cow IDs across frames, even during moderate occlusions or crowding. The tracking speed remained within real-time thresholds, with end-to-end inference and post-processing times averaging 12.8 ms per frame on the Nvidia T4 GPU. Occasional ID switches were noted when multiple cows intersected closely or passed through blind zones, but these did not disrupt aggregate behavior modeling. The tracking pipeline thus proved sufficiently robust for continuous behavior analysis in dense barn environments.

### Re-Identification Model Performance

We successfully ran the re-identification (Re-ID) model in tandem with YOLOv8 and ByteTrack. Due to the single-camera setup, we were only able to train the model to recognize 10 specific cows out of the total 50 cows, using cropped images from detection outputs. In deployment, the model was able to "recognize" six cows, although only one was matched with high visual confidence: a small, light-brown cow that the system tracked consistently.

The others were partially matched based on coat color or pattern but not confirmed as correct. This partial recognition is likely due to most of the trained individuals not being present in the visible frame during tests. The Re-ID model also added memory and computational overhead, which affected the real-time tracking loop, increasing gaps between frame updates, leading to increased time between coordinate data collection to

over 2 seconds. This performance issue, combined with incomplete in-frame representation, led us to exclude the Re-ID model from the current pipeline. We plan to reintegrate it once all five cameras are accessible and the full barn is covered.

```
0: 480x640 4 cows, 11.3ms
Speed: 3.6ms preprocess, 11.3ms inference, 2.0ms postprocess per image at shape (1, 3, 480, 640)
Stopped by user.
Program terminated.
Recognized cattle during session:
- cow_4
- cow_2
- cow_10
- cow_5
- cow_1
- cow_3
(reolinkenv311) C:\Users\lukasmolvaer\Agrointel>
```

Figure 5.2: Terminal output of the re-identification pipeline. The system correctly identifies and maintains unique cow IDs during the session, listing all cows "recognized" before program termination.

## Prediction Model Evaluation

We evaluated our Conv1D prediction model using both hold-out testing and continuous looped predictions to simulate live system behavior.

- **Hold-out Accuracy:** The model achieved an overall accuracy of 88% on the reserved hold-out dataset. Precision and recall scores across the three classes; **healthy**, **in\_heat**, and **sick** were consistently strong, with the highest performance observed in identifying **sick** cows (F1-score of 0.92).

|                                  |             |        |          |         |  |
|----------------------------------|-------------|--------|----------|---------|--|
| Hold-out ⇒ Loss=0.449, Acc=0.881 |             |        |          |         |  |
| 94/94                            | 1s 6ms/step |        |          |         |  |
|                                  | precision   | recall | f1-score | support |  |
| healthy                          | 0.83        | 0.84   | 0.83     | 1001    |  |
| in_heat                          | 0.89        | 0.87   | 0.88     | 999     |  |
| sick                             | 0.92        | 0.93   | 0.92     | 1000    |  |
| accuracy                         |             |        | 0.88     | 3000    |  |
| macro avg                        | 0.88        | 0.88   | 0.88     | 3000    |  |
| weighted avg                     | 0.88        | 0.88   | 0.88     | 3000    |  |

Figure 5.3: Performance on the hold-out test set. The model achieved an overall accuracy of 88%, with balanced F1-scores across the three classes.

- **Batch Variability:** To reflect real-time behavior, we ran repeated batch predictions at runtime. These tests revealed expected fluctuations in accuracy across time windows, with batch F1-scores ranging from 0.77 to 0.90. While some batches exhibited perfect **in\_heat** detection, others showed challenges in identifying **healthy** cows, likely due to synthetic noise or overlap with mild activity signals.

|               |           |        |          |         |
|---------------|-----------|--------|----------|---------|
| Batch report: |           |        |          |         |
|               | precision | recall | f1-score | support |
| healthy       | 0.94      | 0.80   | 0.86     | 20      |
| in_heat       | 1.00      | 0.94   | 0.97     | 18      |
| sick          | 0.75      | 1.00   | 0.86     | 12      |
| accuracy      |           |        | 0.90     | 50      |
| macro avg     | 0.90      | 0.91   | 0.90     | 50      |
| weighted avg  | 0.92      | 0.90   | 0.90     | 50      |

Figure 5.4: Prediction performance on a runtime batch. The model achieved an accuracy of 90% in this window, with strong classification of `in_heat` and `sick` cows.

|               |           |        |          |         |
|---------------|-----------|--------|----------|---------|
| Batch report: |           |        |          |         |
|               | precision | recall | f1-score | support |
| healthy       | 0.79      | 0.58   | 0.67     | 19      |
| in_heat       | 0.75      | 0.88   | 0.81     | 17      |
| sick          | 0.81      | 0.93   | 0.87     | 14      |
| accuracy      |           |        | 0.78     | 50      |
| macro avg     | 0.78      | 0.80   | 0.78     | 50      |
| weighted avg  | 0.78      | 0.78   | 0.77     | 50      |

Figure 5.5: Prediction performance on a second runtime batch. Although overall accuracy dropped to 78%, the model maintained high recall for `sick` and `in_heat` categories. This illustrates the dynamic nature of time-series prediction depending on input behavior segments.

- **Model Generalization:** Despite training on synthetic labels, the model maintained stable generalization performance across unseen input samples. The convolutional architecture efficiently captured short-term temporal dependencies in behavior sequences without overfitting.
- **Evaluation Strategy:** Performance metrics were calculated using macro and weighted averages of precision, recall, and F1-score, offering a balanced view of per-class reliability. We also monitored performance across multiple batches to assess prediction consistency.

These results demonstrate the model’s capacity to distinguish behavioral categories based on structured input features. While grounded in synthetic data, this evaluation validates our architecture and processing pipeline, setting a solid foundation for further real-world deployment and supervised fine-tuning with labeled veterinary data.

## 5.2 Relevance to Research Literature

Our results align closely with recent advancements in AI-driven livestock monitoring. High detection accuracy achieved using YOLOv8 supports prior research demonstrating the effectiveness of YOLO-based models in barn environments. For instance, Hao et al. [5] and Zheng et al. [6] highlight YOLOv5 and YOLO-BYTE as powerful tools for robust cattle detection in noisy, real-world settings.

Our integration of ByteTrack for multi-animal tracking is also consistent with best practices outlined in Han et al. [7], who found it well-suited to crowded, occlusion-prone farm environments. The core tracking algorithm (ByteTrack) is recognized for its ability to handle missed detections while maintaining ID continuity [8].

Compared to existing literature, our implementation goes further by developing an end-to-end, real-time pipeline that spans detection, tracking, behavioral analysis, and health prediction. While previous works (e.g., Distante et al. [3]) provide meta-reviews of individual AI modules in livestock farming, few demonstrate a unified, deployable system in active use. Our deployment in a Norwegian winter barn helps close this gap.

Furthermore, our behavioral prediction approach; linking reduced activity to illness and increased movement to estrus, is supported by sensor-based findings in smart farming literature [15]. Arıkan et al. [12] also demonstrated that estrus can be detected using deep learning applied to behavioral sequences, confirming the value of time-series analysis for reproductive monitoring.

Although our predictive model is currently trained on synthetic labels, its output shows patterns that reinforce real-world observations. Myint et al. [14] similarly emphasize that early signs of health issues such as lameness can be detected via motion anomalies, even using a single camera.

By incorporating these findings into a functioning pipeline, our project substantiates the applicability of recent AI research in practical barn settings and offers a foundation for future studies to build upon with labeled veterinary data.

## 5.3 Benefits for the Client

The system delivers clear benefits to Agroidintel:

- **Real-time cattle monitoring:** Provides continuous updates on cow locations and zone presence.
- **Behavior summaries:** Offers easy-to-interpret snapshots that highlight activity and rest trends.
- **Predictive alerts:** Early warnings for possible health issues or heat events can reduce veterinary costs and improve breeding efficiency.
- **Cloud integration:** The use of Firebase allows data to be accessed remotely and shared with other services or dashboards.

In addition, the system enables Agroidintel to collect a growing behavioral dataset. This data can power future analytics features, like herd comparison, trend detection, and more personalized animal care strategies. The delivery of this working prototype also demonstrates Agroidintel’s technical capability to partners and stakeholders.



## 5.4 Summary

This chapter evaluated the system across detection, tracking, re-identification, and behavioral prediction. Our YOLOv8 model achieved high precision and recall, confirming its reliability in barn conditions. ByteTrack maintained consistent tracking performance with minimal identity switches. While the re-identification model showed early promise, limited visibility and compute overhead led to its temporary exclusion from the real-time loop.

The Conv1D prediction model demonstrated stable classification performance on synthetic data, validating the feasibility of using behavioral features for health and estrus detection.

Overall, the system meets its intended goals for object tracking and behavioral analysis, while laying the groundwork for future health prediction with real veterinary labels.

# Chapter 6

## Ethical Considerations

### Improving Animal Welfare

The core ethical goal of this system is to support better animal welfare. By monitoring cattle movement and zone activity in real time, the system can detect early signs of health issues or estrus. Timely alerts enable prompt veterinary attention and improved breeding management. This proactive approach reduces animal suffering and enhances the productivity and wellbeing of the herd.

### Accuracy and Transparency

Any AI system carries a risk of incorrect outputs. False positives could lead to unnecessary interventions, while false negatives might cause important issues to go unnoticed. Our system mitigates this by accompanying alerts with interpretable reasoning (e.g., "low movement for 6 hours"). Farmers are encouraged to treat alerts as indicators rather than absolute diagnoses and to verify findings through observation or review of the camera footage.

We have prioritized transparency in the system's logic. Instead of providing opaque alerts, each health or estrus warning is traceable to specific behavioral data, such as distance moved, resting time, or transition frequency. This design helps build trust with users and ensures that they understand why the system is making a recommendation.

### Privacy and Consent

While our system is focused on cattle, humans (e.g., farm staff) may appear in camera footage. The model only labels them generically as "person" and does not attempt to identify or analyze human behavior. We recommend informing farm personnel about the presence of video monitoring and ensuring that all data is used solely for animal management. No video is transmitted externally unless explicitly configured by the client.

### Respecting Farmer Expertise

AI should augment, not replace, human decision-making. Our system handles tedious observation tasks, freeing up time for farmers to focus on direct care and more meaningful work. We worked closely with our client during development, regularly incorporating feedback from the field to ensure the system fits real-world farming workflows. This

participatory approach respects the expertise of farmers and keeps them at the center of decision-making.

## **6.1 Summary**

The system is well-suited for deployment on real farms. Its modular design supports flexibility and ease of maintenance. Ethically, it improves animal welfare, preserves transparency, protects privacy, and empowers farmers. Our approach ensures that the system acts as a reliable and responsible assistant in the evolving landscape of smart farming.

# Chapter 7

## Reflections

### 7.1 Technical and Personal Growth

Embarking on this project provided an invaluable opportunity to apply theoretical knowledge to a practical, real-world problem in the agricultural domain. The integration of computer vision, machine learning, and cloud-based technologies to monitor and analyze cattle behavior challenged us to think critically and adaptively.

We began by exploring object detection methodologies, selecting YOLOv8 for its balance between speed and accuracy in real-time applications [16]. Implementing YOLOv8 required a deep understanding of convolutional neural networks and the nuances of training models on specialized datasets. Transitioning to object tracking, we employed ByteTrack, which allowed for robust multi-object tracking by associating every detection box [17]. This phase honed our skills in handling temporal data and understanding the complexities of maintaining object identities across frames.

Data management was facilitated through Firebase, a cloud-based platform that enabled real-time data synchronization and storage [18]. Integrating Firebase into our system architecture taught us the importance of scalable and reliable backend solutions, especially when dealing with continuous data streams.

Throughout the project, we developed and refined several key skills:

- **Data Processing:** Handling and preprocessing large volumes of video data, including frame extraction, annotation, and augmentation.
- **Model Training and Evaluation:** Training machine learning models/deep learning layers with limited labeled data, employing techniques such as transfer learning and cross-validation to enhance performance.
- **System Integration:** Combining various components; object detection, tracking, data storage, and analysis, into a cohesive and functional pipeline.
- **Cloud Computing:** Utilizing cloud services for data storage and real-time updates, ensuring scalability and accessibility.
- **Visualization:** Creating intuitive visual representations of data and model outputs to facilitate interpretation and decision-making.

This journey not only expanded our technical repertoire but also instilled a deeper appreciation for the interdisciplinary nature of modern technological solutions.

## 7.2 Challenges and Lessons Learned

The path to developing a functional and reliable cattle monitoring system was full of challenges that tested our problem-solving abilities and resilience.

One significant hurdle was ensuring the accuracy of object detection and tracking under varying environmental conditions. Factors such as lighting changes, occlusions, and the presence of multiple similar-looking objects often led to detection errors. We addressed these issues by implementing data augmentation techniques during training and fine-tuning model parameters to improve robustness.

Integrating multiple system components posed another challenge. Ensuring seamless communication between the detection module, tracking algorithm, and the Firebase backend required meticulous planning and testing. We learned the importance of modular design and the use of standardized data formats to facilitate interoperability.

Data synchronization was also a concern, particularly in maintaining consistency between real-time video feeds and the corresponding data entries in Firebase. Implementing time-stamping mechanisms and rigorous validation checks helped mitigate discrepancies.

These challenges underscored several key lessons:

- **Adaptability:** The ability to adjust methodologies in response to unforeseen obstacles is crucial in project development.
- **Attention to Detail:** Small oversights can lead to significant issues; thorough testing and validation are essential.
- **Effective Communication:** Clear documentation and communication among team members streamline the development process and reduce errors.
- **Continuous Learning:** Staying updated with the latest research and tools can provide innovative solutions to complex problems.

## 7.3 Teamwork and Collaboration

This project was officially carried out by a team of two data science students: Lukas and Ahaan. Although we shared the same academic background and end goals, our involvement in the project unfolded at different times and in different capacities.

In the first phase of the project, Lukas worked independently for several months to establish the system's technical foundation. This included setting up the on-site camera infrastructure, collecting and annotating training data, training the YOLOv8 object detection model, integrating it with the ByteTrack tracking algorithm, developing the behavioral feature extraction pipeline, and conducting early experimentation with a re-identification model. This foundational work laid the groundwork for the system's ability to operate in real-time and generate meaningful behavioral data.

Ahaan became actively involved during the second half of the project. He was responsible for implementing the initial version of the prediction model using time-series behavioral features. In addition to his modeling contributions, Ahaan also played a significant role in writing, organizing, and editing the final report. He helped structure several chapters, added relevant literature, ensured consistency across sections, and linked code documentation to the report.

When we began collaborating more closely during the later stages of the project, our teamwork proved highly effective. We coordinated tasks using Google Docs and Overleaf for real-time co-authoring, and email and shared cloud drives for code review and debugging. This phase of the project demonstrated our ability to align perspectives, divide responsibilities clearly, and iterate quickly despite having different working rhythms earlier in the process.

Through this collaboration, we gained experience in:

- Coordinating across different stages of a long-term project with evolving team dynamics.
- Aligning distinct technical strengths to create a cohesive final product.
- Maintaining clear, respectful communication in both asynchronous and real-time collaboration settings.

# Chapter 8

## Conclusion

### 8.1 Summary of Achievements

This bachelor project set out to develop an AI-based cattle monitoring system capable of detecting, tracking, and analyzing cattle behavior in a winter barn environment. Over the course of the project, we successfully designed and implemented a full end-to-end pipeline. This pipeline includes:

- A YOLOv8-based object detection model capable of identifying cows and humans in real-time with high precision.
- Integration of the ByteTrack algorithm for maintaining consistent cow identities across frames, with zone-based tracking.
- A Firebase-backed data infrastructure to store raw tracking data, zone entries, activity summaries, and prediction outputs.
- An analytical module that computes hourly and cumulative behavior features from coordinate and zone data.
- A 1D convolutional neural network (Conv1D) model trained on synthetic labels to predict cow health and estrus status.
- Early experimentation with a re-identification (Re-ID) model to support consistent cow ID assignment across occlusions and reentries.

These components function together in real-time and demonstrate the feasibility of using computer vision for autonomous livestock behavior monitoring. The system can generate structured insights from raw barn footage, offering clear value to farmers and animal welfare monitoring.

### 8.2 Degree of Goal Achievement

Our project defined three primary goals at the outset:

1. **Robust Object Detection:** Fully achieved. The YOLOv8 model trained on our annotated dataset reached strong performance metrics on cow and human detection, validated both qualitatively and quantitatively.

2. **Reliable Tracking System:** Achieved. ByteTrack, combined with our defined zone system, allowed us to maintain individual cow tracking throughout live video feeds.
3. **Predictive Behavior Analysis:** Partially achieved. We implemented a functioning prediction model using Conv1D and a well-structured set of engineered features. However, real-world health labels were unavailable, so the model had to be trained on simulated labels. Despite this, evaluation results suggest strong potential.

In addition, our re-identification (Re-ID) experiments yielded mixed but informative results. The model successfully recognized a subset of trained cows, albeit with performance degradation due to limited camera coverage and increased computational overhead. Re-ID has been excluded from production use for now but is positioned for reintegration once multi-camera support becomes available.

## 8.3 Future Work

Several areas were identified for further development and expansion:

- **Full Multi-Camera Integration:** Unlocking all five barn cameras would dramatically improve tracking consistency and eliminate ID resets. It would also enable better Re-ID model training and validation.
- **Direct RTSP Cloud Access:** Completing Reolink Open Platform (ROP) integration would allow direct, programmatic access to camera feeds and eliminate the need for screen recording.
- **Real-Label Health Data Collection:** Ground-truth health and estrus labels from the farm would enable supervised learning and significantly enhance the accuracy and credibility of the predictive model.
- **Improved Re-ID and Model Fusion:** With broader camera coverage, Re-ID models can support cross-frame and cross-camera identity preservation. Lightweight architectures will need to be explored to maintain real-time performance.
- **Farmer-Facing Dashboard:** Integrating our Firebase data into a responsive dashboard (mobile or web) would allow real-time alerts, visualizations, and reports to assist farm decision-making.
- **Sensor Fusion:** Combining vision-based data with IoT sensor data (e.g., temperature, RFID, or milk output) could refine behavior analysis and provide a more holistic health monitoring framework.
- **Deployment as a SaaS Tool:** With cloud scaling, the system could support multiple barns or farms simultaneously. Each could be isolated in the database, with client-specific configurations.

This project serves as a proof of concept for smart livestock monitoring and lays a technical foundation. Future iterations and pilot deployments will determine how this prototype can be adapted into a full commercial product. The system’s modular design ensures it can evolve as new capabilities and data become available.



## 8.4 Conclusion

This project has been a transformative experience, bridging the gap between academic knowledge and practical application. Over several months of intensive development, experimentation, and iteration, we have not only built a technically functional system but also gained a profound understanding of how interdisciplinary tools-computer vision, machine learning, and cloud computing-can converge to solve real-world agricultural challenges.

From managing complex pipelines to resolving unexpected bugs in cloud infrastructure, every challenge demanded a blend of creativity, persistence, and teamwork. Beyond technical learning, we cultivated a problem-solving mindset, resilience under uncertainty, and the ability to balance ambitious goals with practical limitations.

Our achievements include developing an accurate real-time cattle monitoring system capable of detecting individual cows, tracking their behavior, and providing actionable health predictions. While the system currently relies on synthetic training data and a single-camera setup, addressing these limitations by integrating multiple camera streams and collecting real-world health data would significantly enhance its accuracy and robustness.

Looking ahead, this experience will serve as a foundation for tackling larger and more intricate projects. It has reinforced our belief that AI systems, when designed thoughtfully, can bring meaningful value to traditional sectors. While this bachelor project marks the end of our academic journey at Høyskolen Kristiania, it also signals the beginning of a lifelong engagement with responsible and impactful innovation.

We leave this project with not only a functioning prototype but with a sense of accomplishment, curiosity, and confidence to continue building solutions that matter.

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# Appendix A

## Link To Major Code Files

The following GitHub repository contains all the code that played a major and necessary role in our project pipeline. The included files are:

- ‘AgroIntel.YOLOV8TrainModel’ Trains a YOLOv8 object detection model to identify and localize cows in barn video frames.
- ‘Detection\_Tracking’ Performs real-time detection and tracking of cows using the trained YOLOv8 model and ByteTrack algorithm.
- ‘AgroIntel.Feature\_Extraction\_Analysis’ Extracts behavioral and movement features such as speed, stops, and zone transitions from tracked cow data for further analysis.
- ‘AgroIntel.Prediction\_Model’ Uses extracted features to train and test a machine learning model that predicts potential sickness or estrus in cows.
- ‘Train\_reID’ Trains the reID-model and saves the best weights to be used for inference.
- ‘Tracking\_reID’ runs the reID-model we trained in “Train\_reID.py” in combination with the “Detection\_Tracking.py” script. Its essentially identical to “Detection\_Tracking.py”, only including the reID-model.

Link: GitHub Repository: AgroIntel Cattle Monitoring  
<https://github.com/ahsarks/AgroIntelBPCodeFiles>

# Appendix B

## Additional Figures and Tables

This is a sample of the graphs and plots we derive from the activity data. We do create plots visualizing both total and daily activity for each cow. However, because we only have about four hours of coordinate and activity data, the “overall” plots meant to show activity levels over multiple days do not provide much insight for now and only show a dot on most graphs. We did not include these graphs here. We do gather plots for every single cow, but we have only included individual graphs for cow 10 and 11 in this sample.

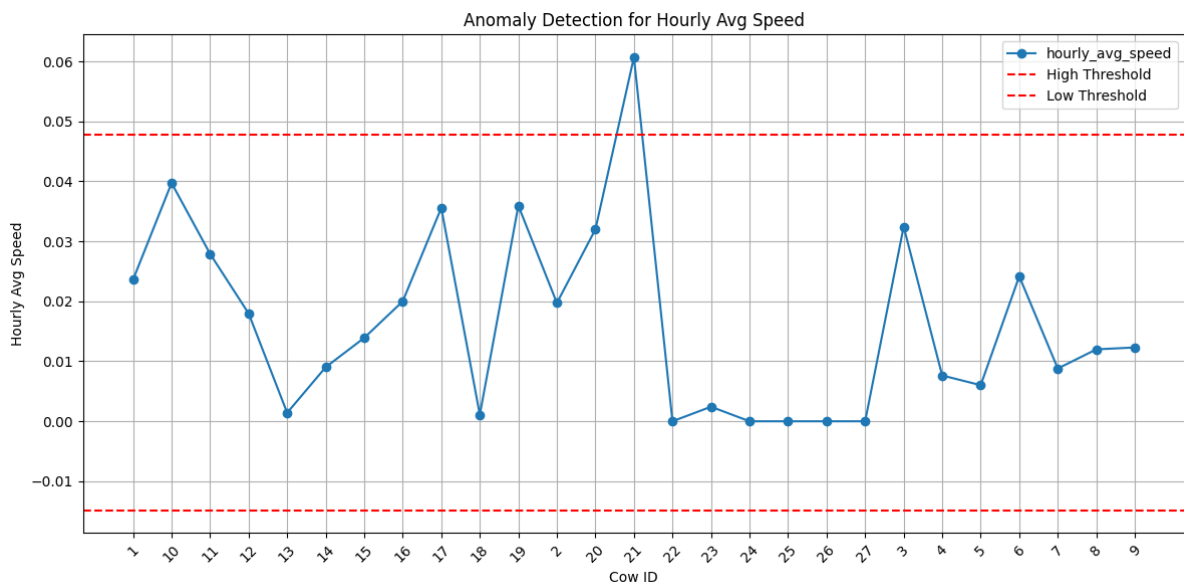


Figure B.1: anomaly\_hourly\_avg\_speed

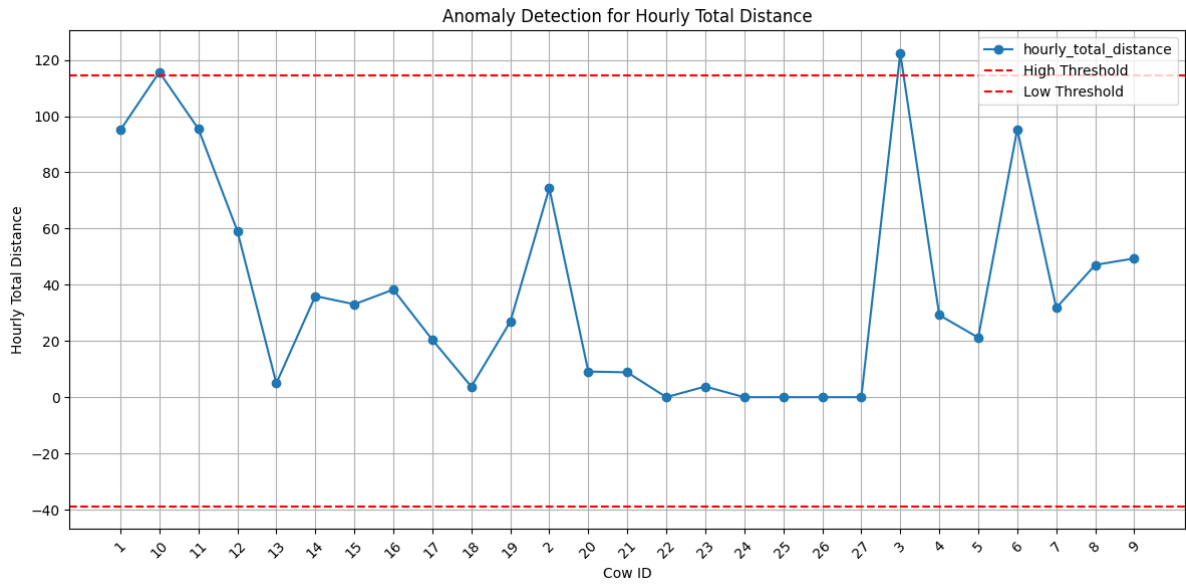


Figure B.2: anomaly\_hourly\_total.distance

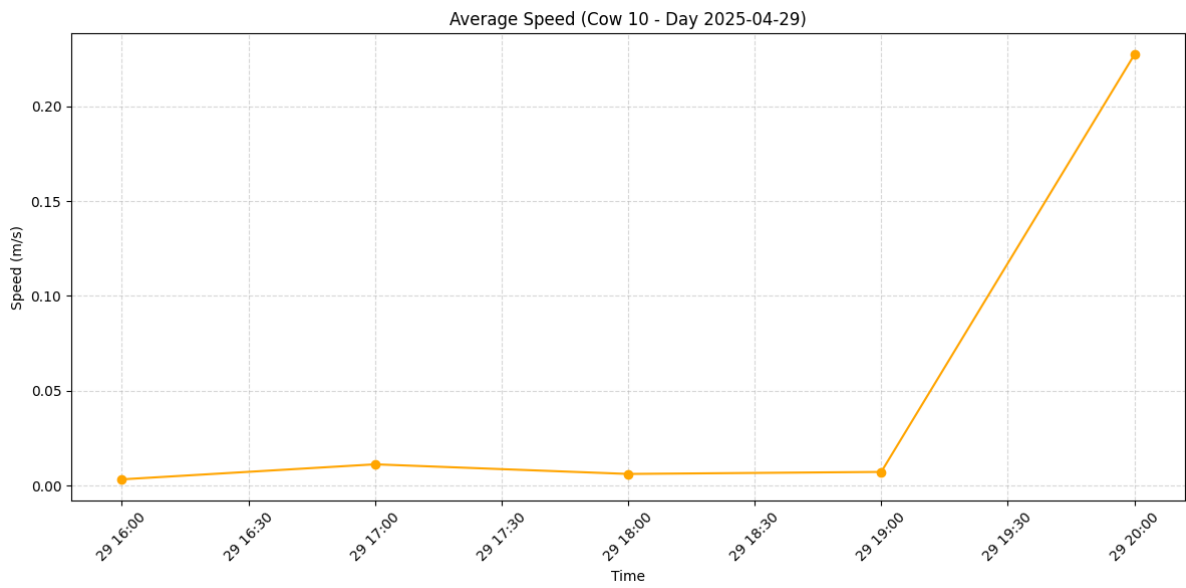


Figure B.3: avg\_speed\_cow\_10\_day

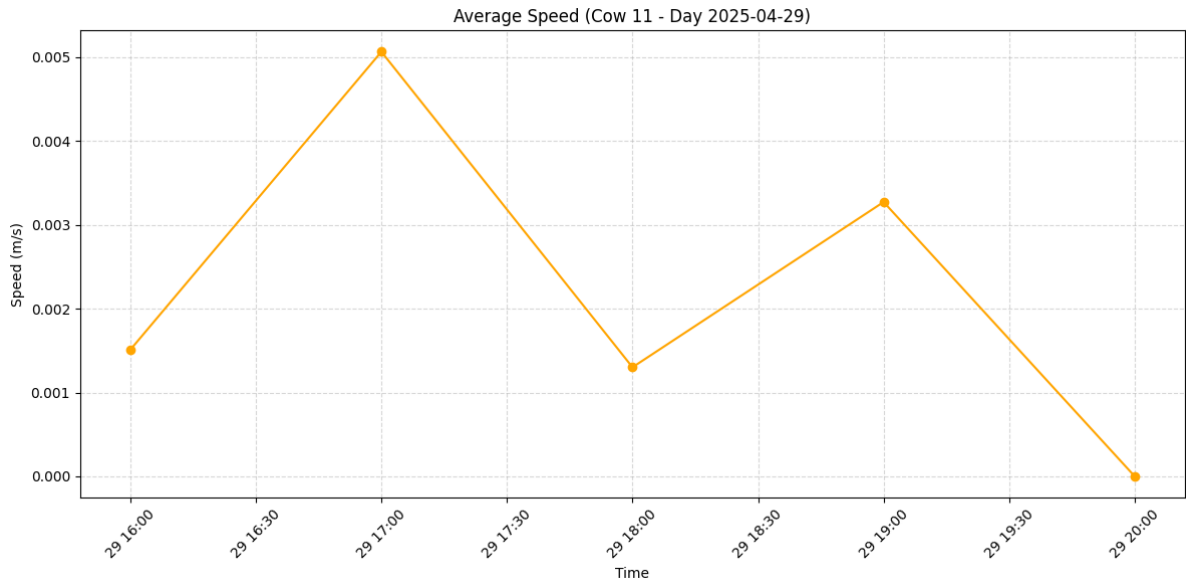


Figure B.4: avg\_speed\_cow\_11\_day

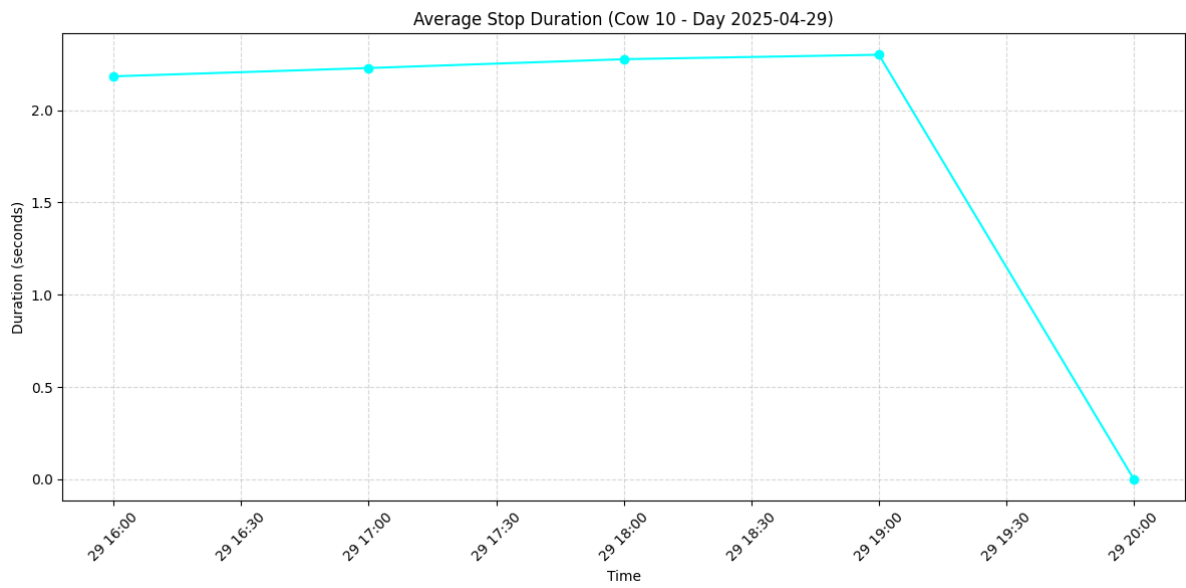


Figure B.5: avg\_stop\_duration\_cow\_10\_day

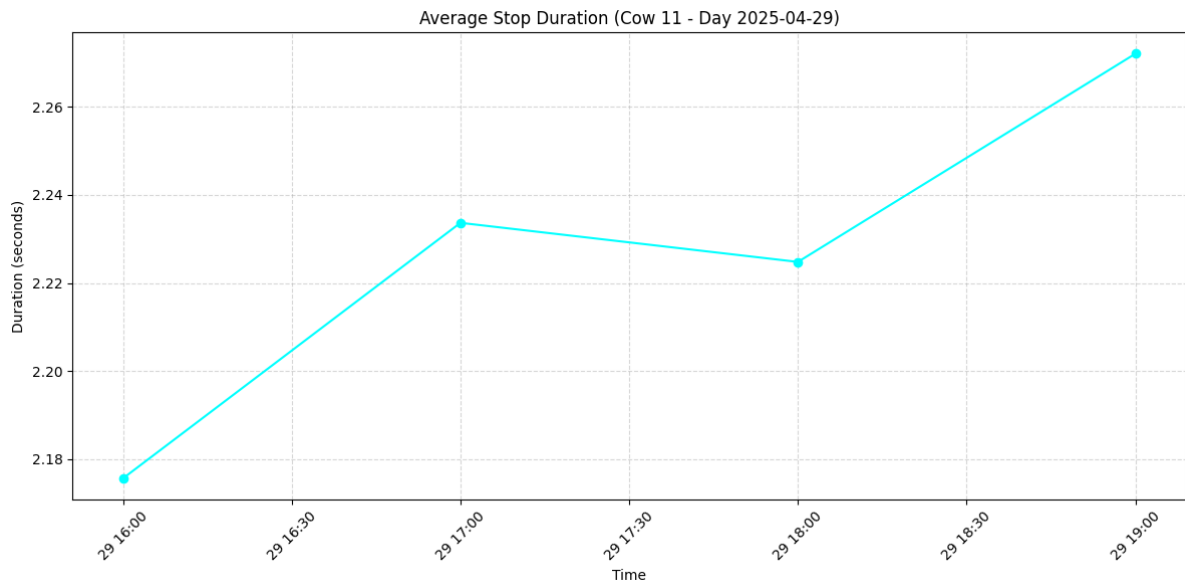


Figure B.6: avg\_stop\_duration\_cow\_11\_day

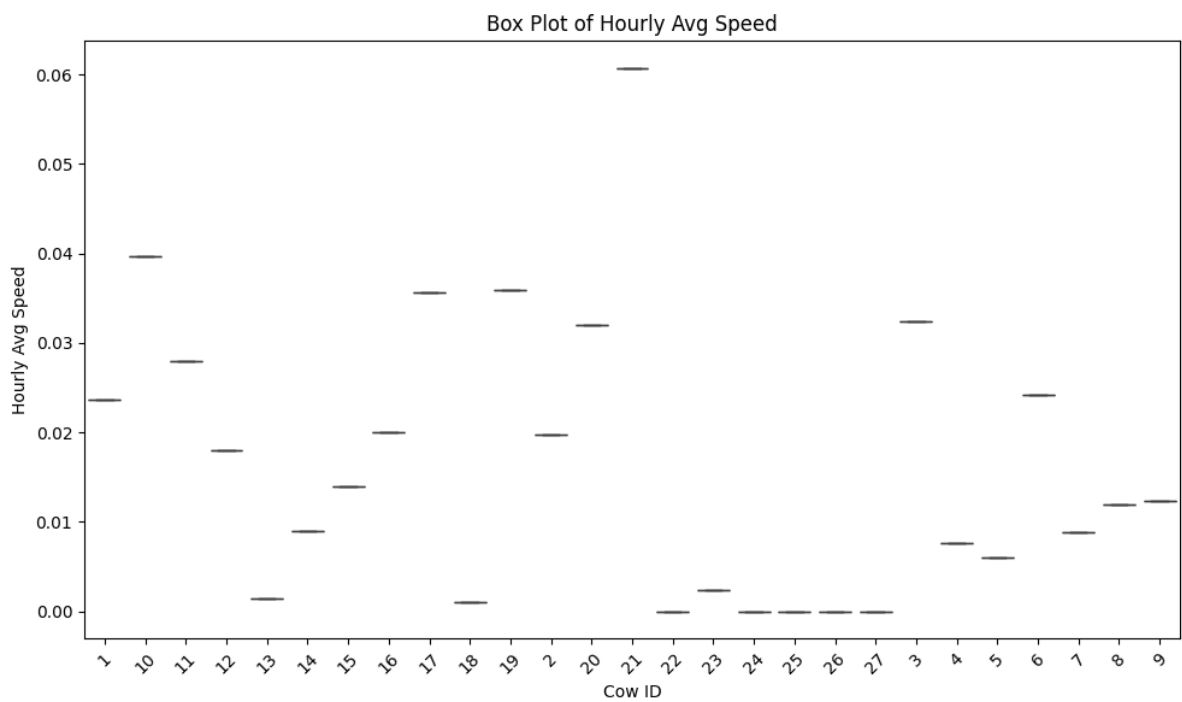


Figure B.7: boxplot\_hourly\_avg\_speed



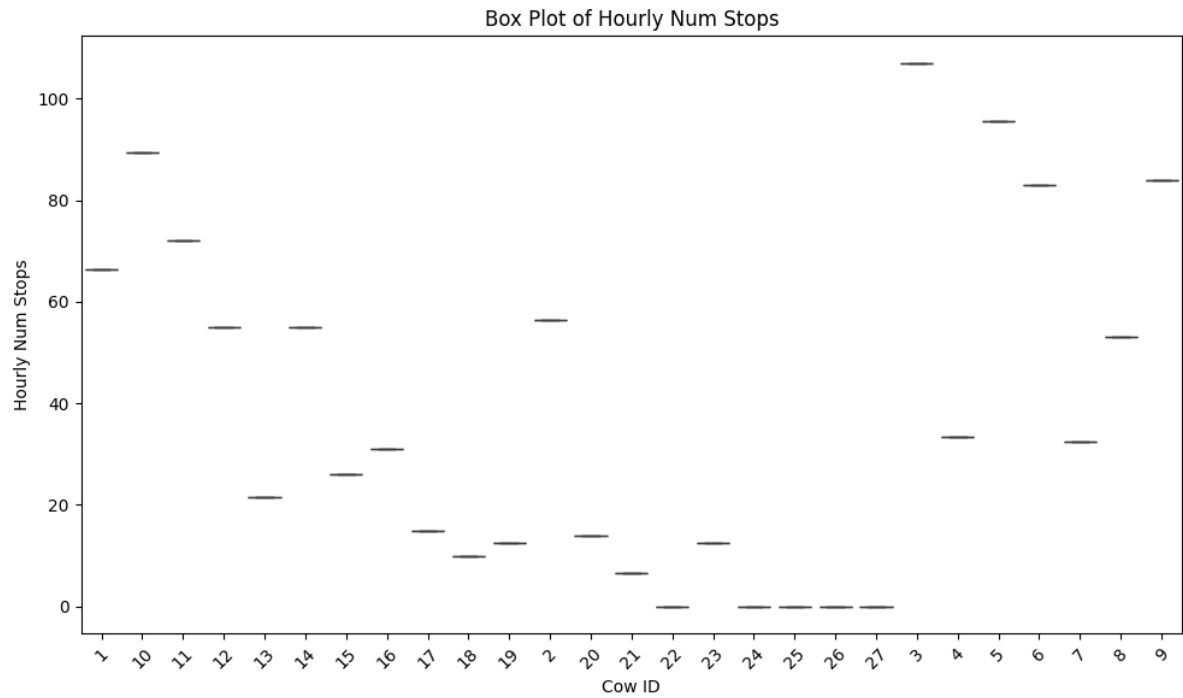


Figure B.8: boxplot\_hourly\_num\_stops

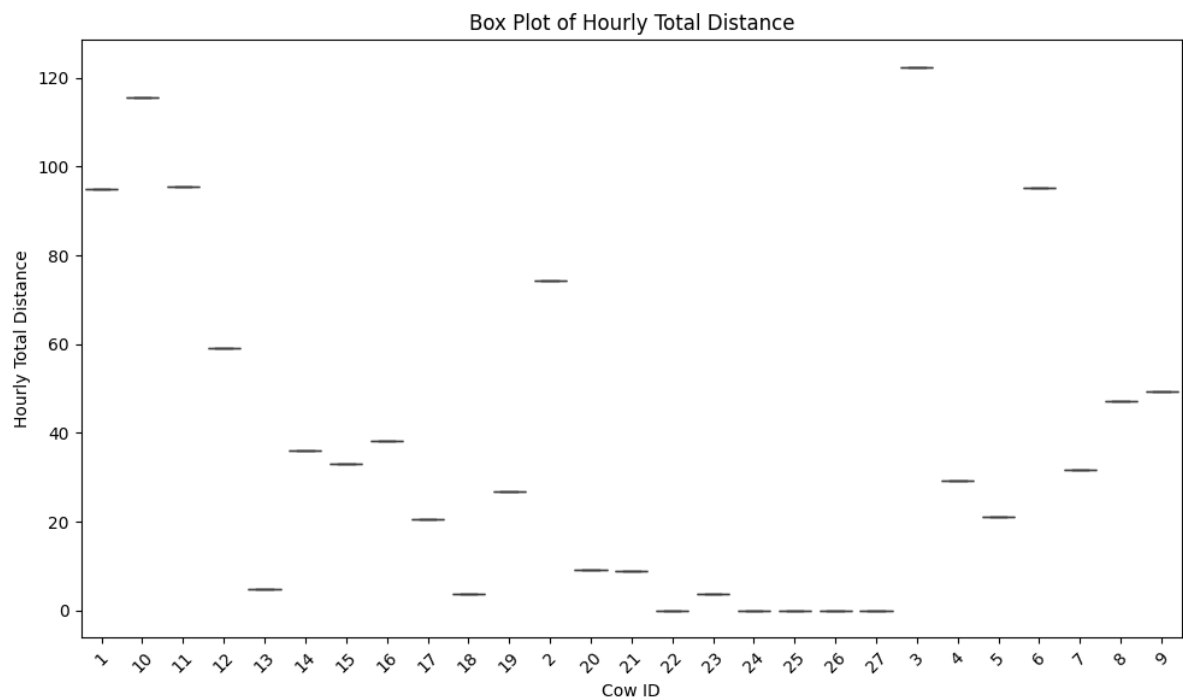


Figure B.9: boxplot\_hourly\_total\_distance

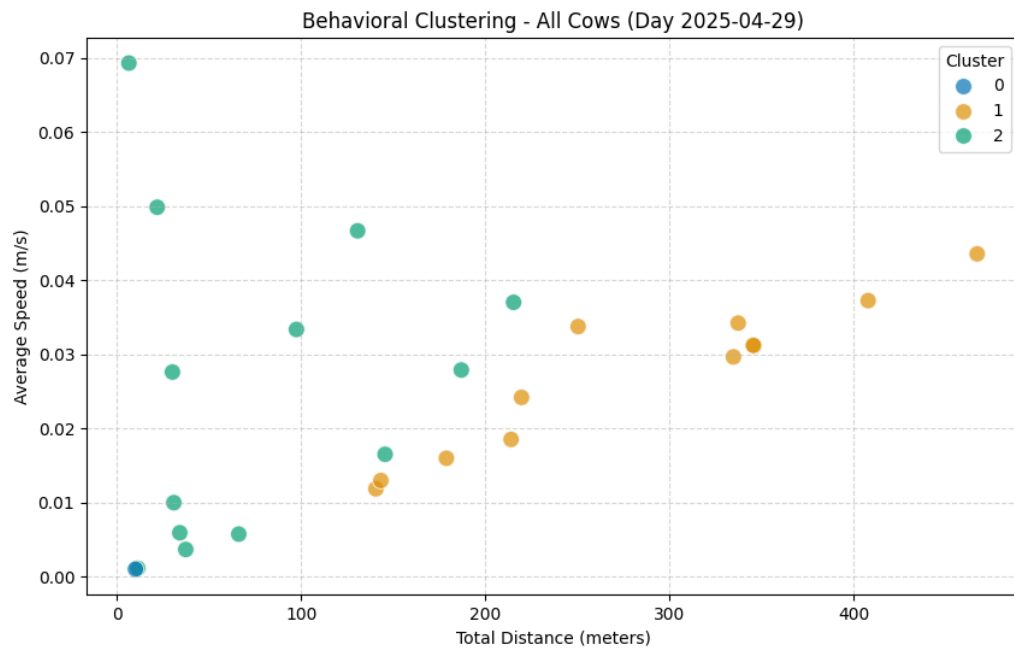


Figure B.10: clustering\_day

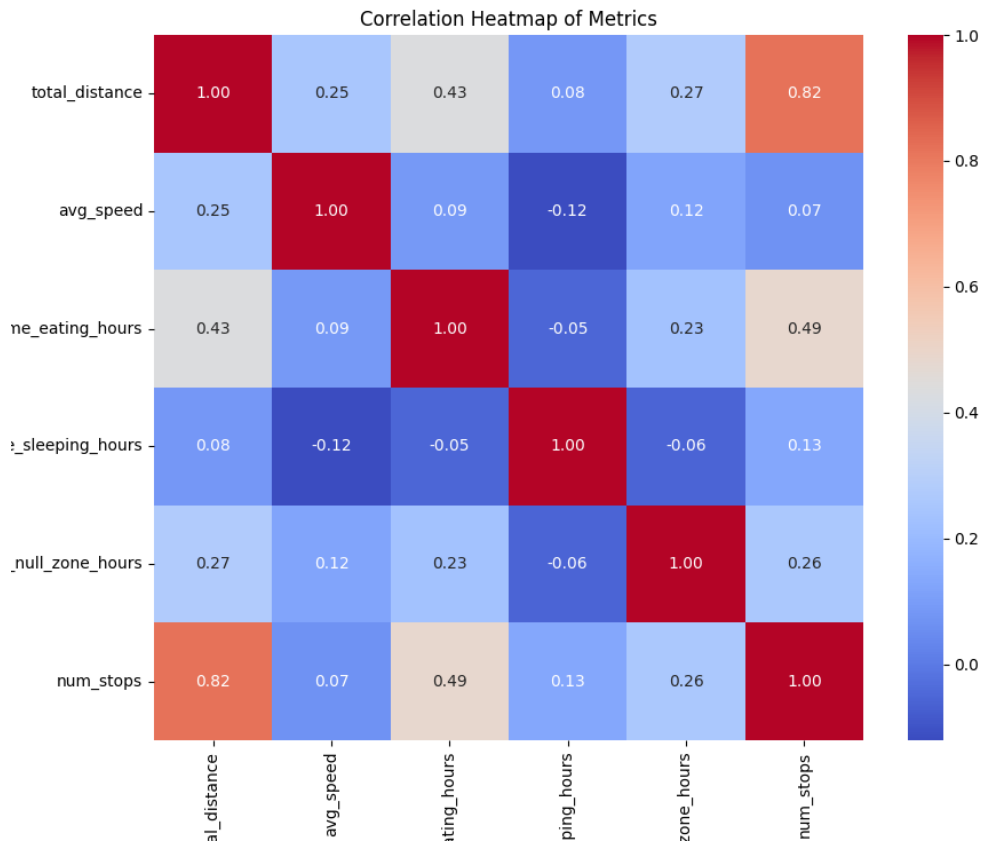


Figure B.11: correlation\_heatmap

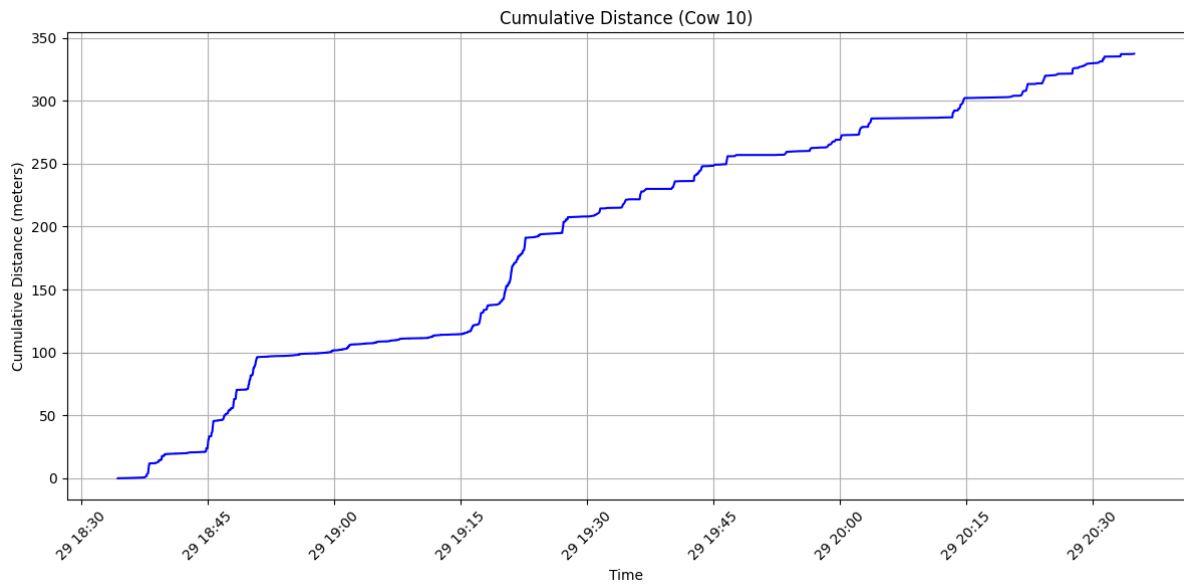


Figure B.12: cumulative\_distance\_cow\_10

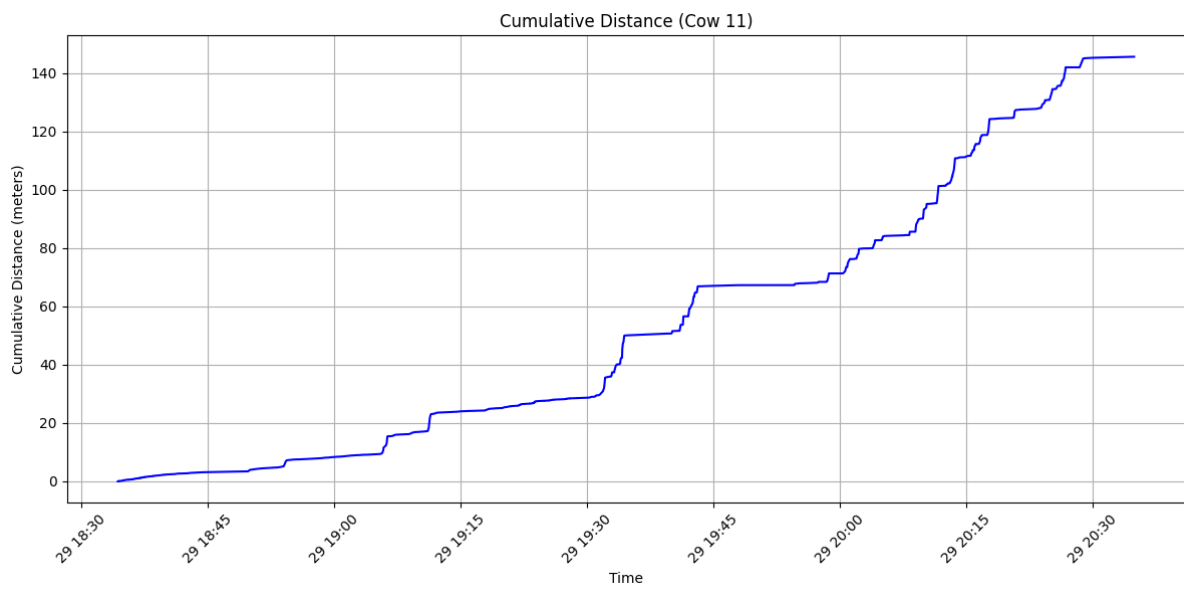


Figure B.13: cumulative\_distance\_cow\_11

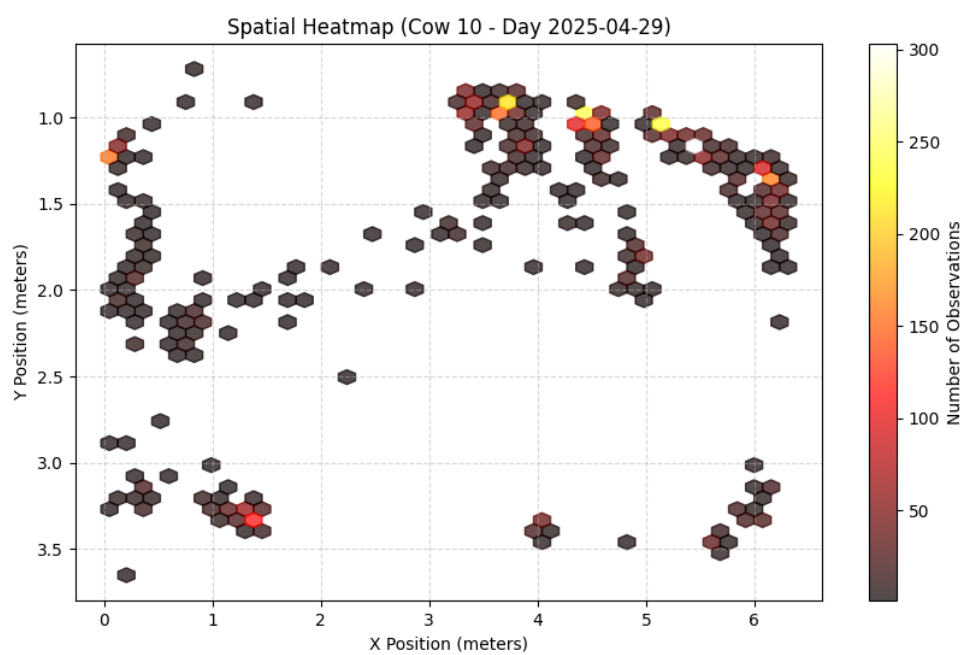


Figure B.14: heatmap\_cow\_10\_day

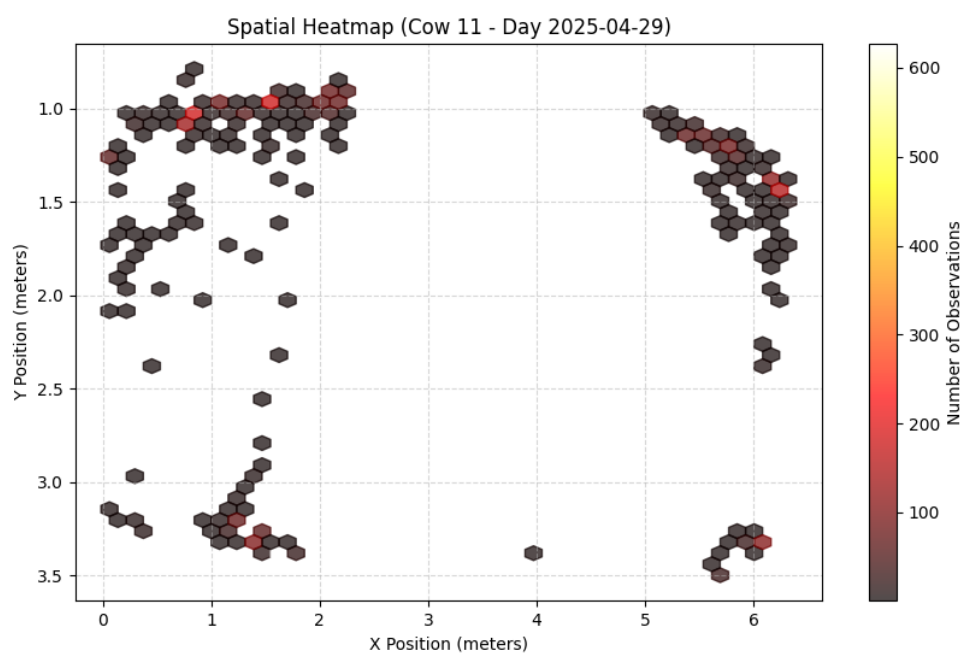


Figure B.15: heatmap\_cow\_11\_day

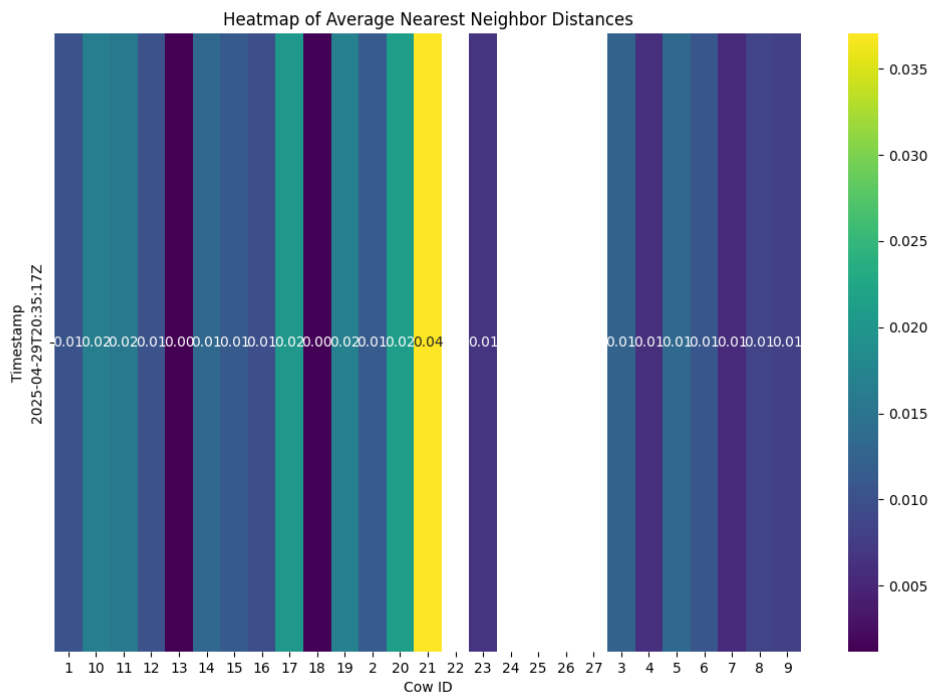


Figure B.16: interaction\_heatmap

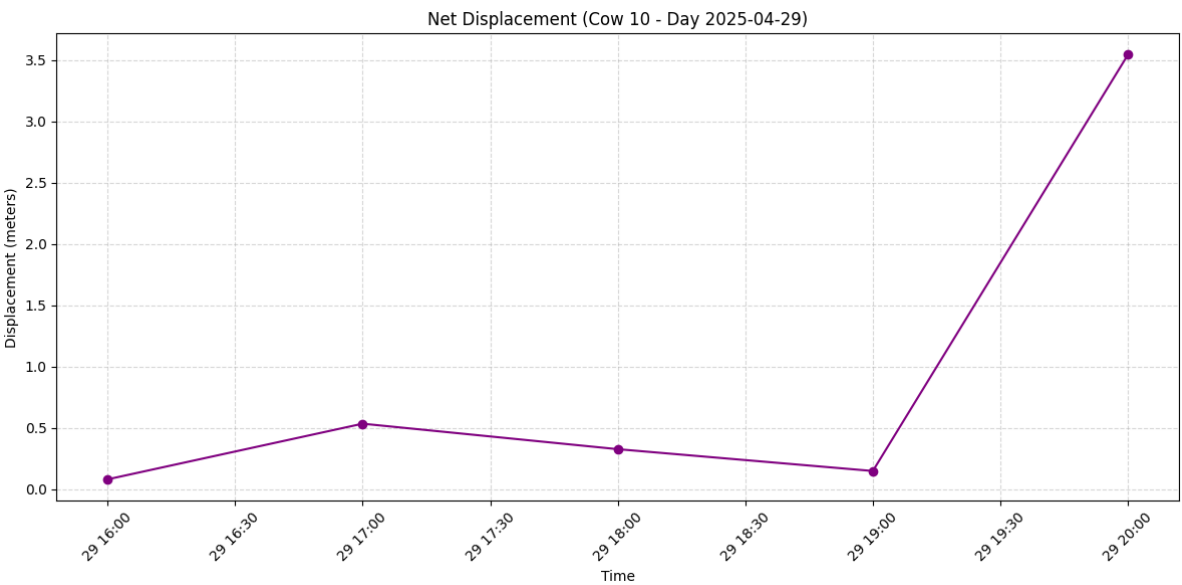


Figure B.17: net\_displacement\_cow\_10\_day

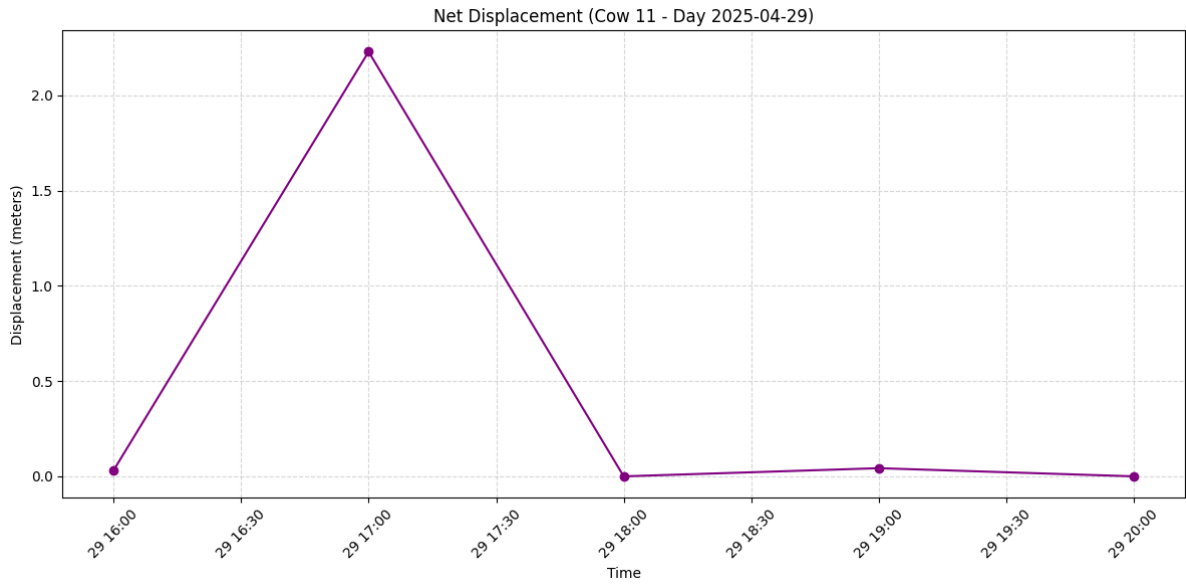


Figure B.18: net\_displacement\_cow\_11\_day

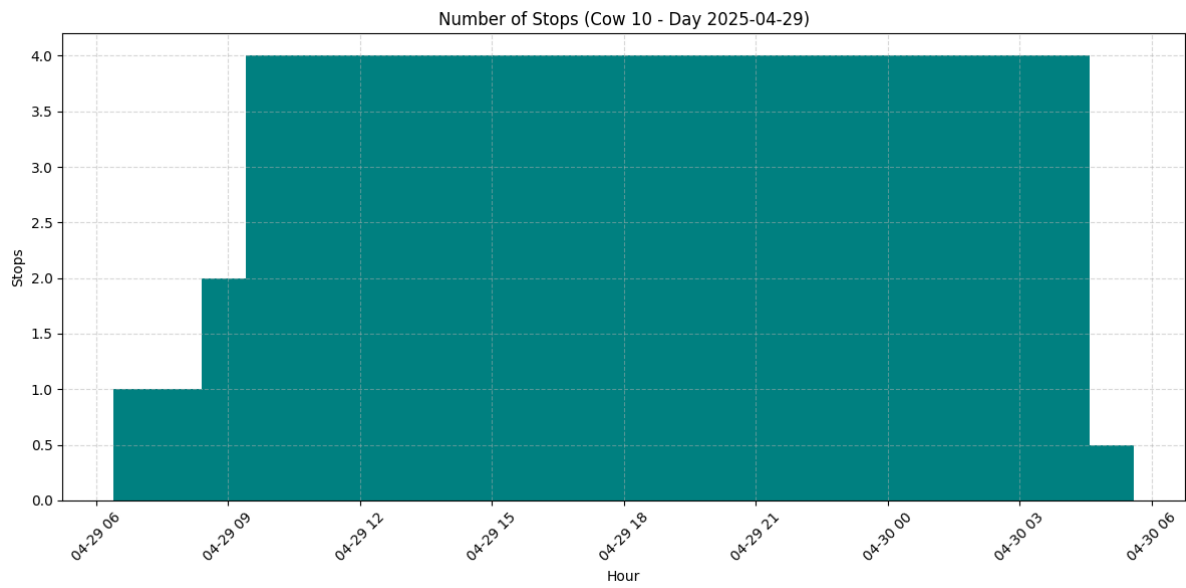


Figure B.19: num\_stops\_cow\_10\_day

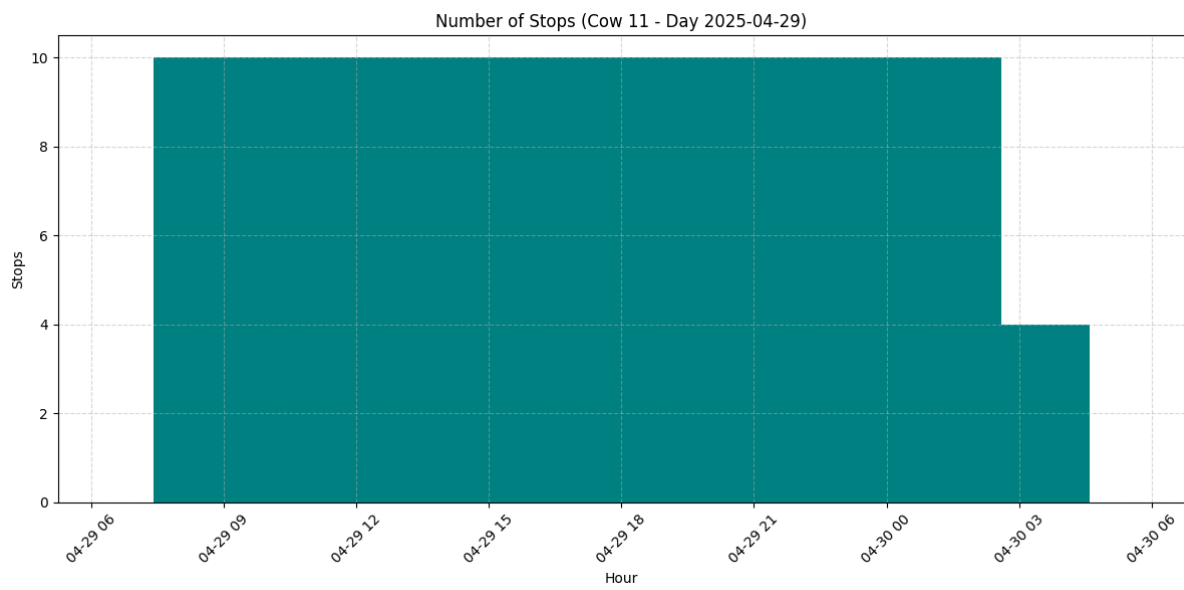


Figure B.20: num\_stops\_cow\_11\_day

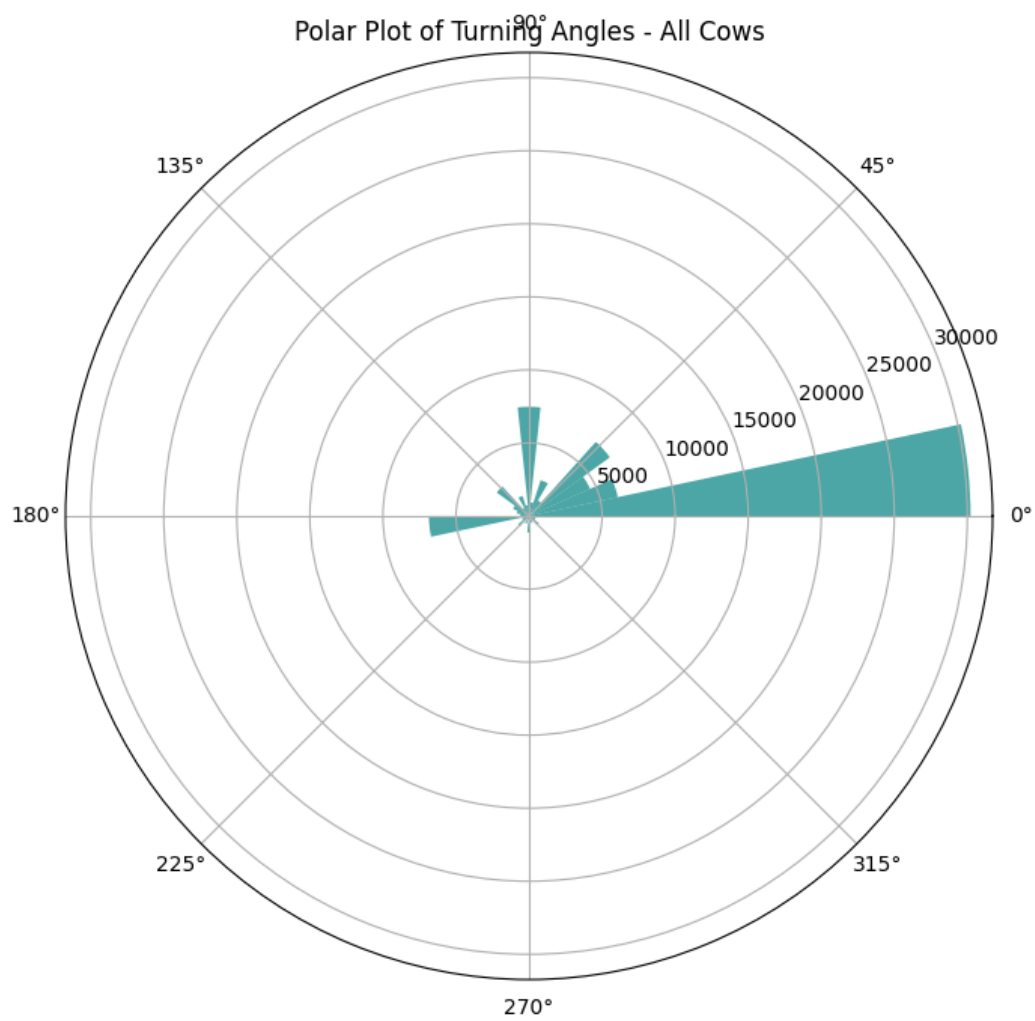


Figure B.21: polar\_turning\_angles\_all



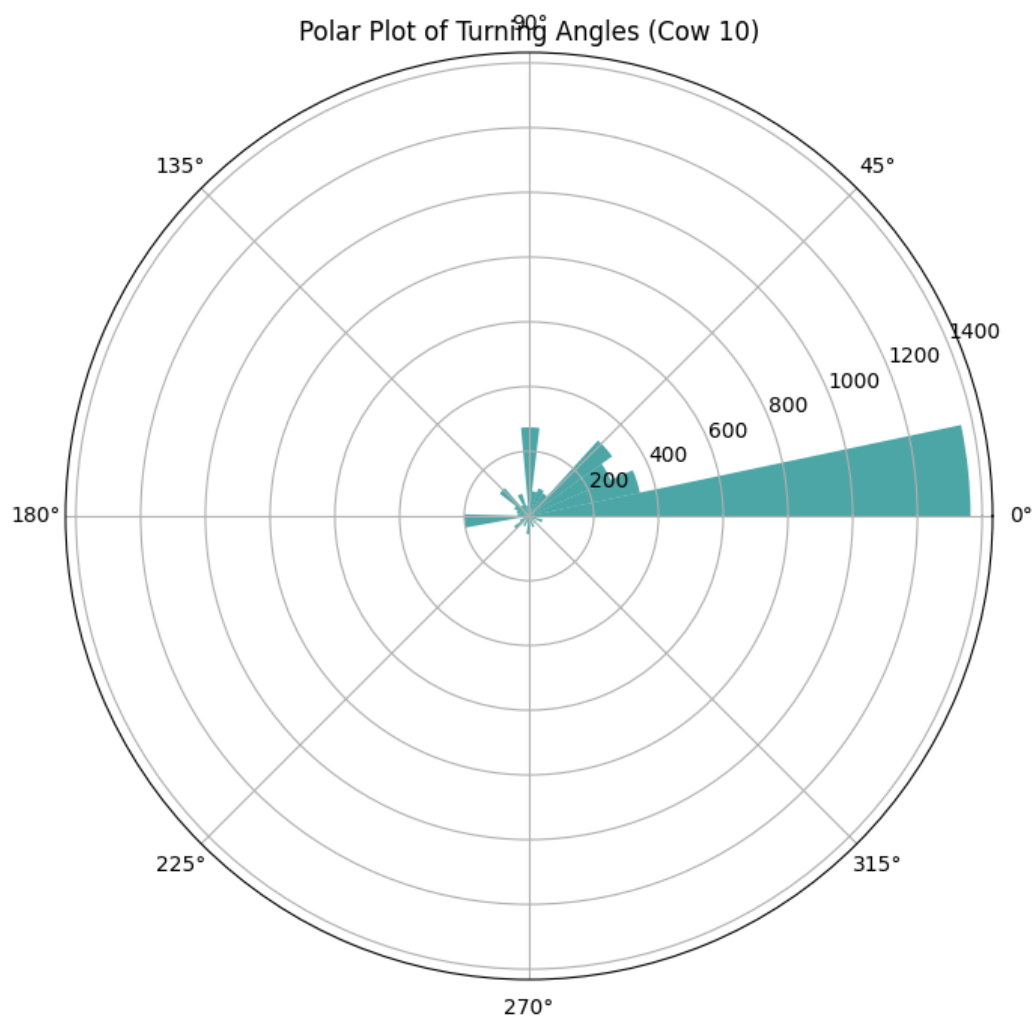


Figure B.22: polar\_turning\_angles\_cow\_10

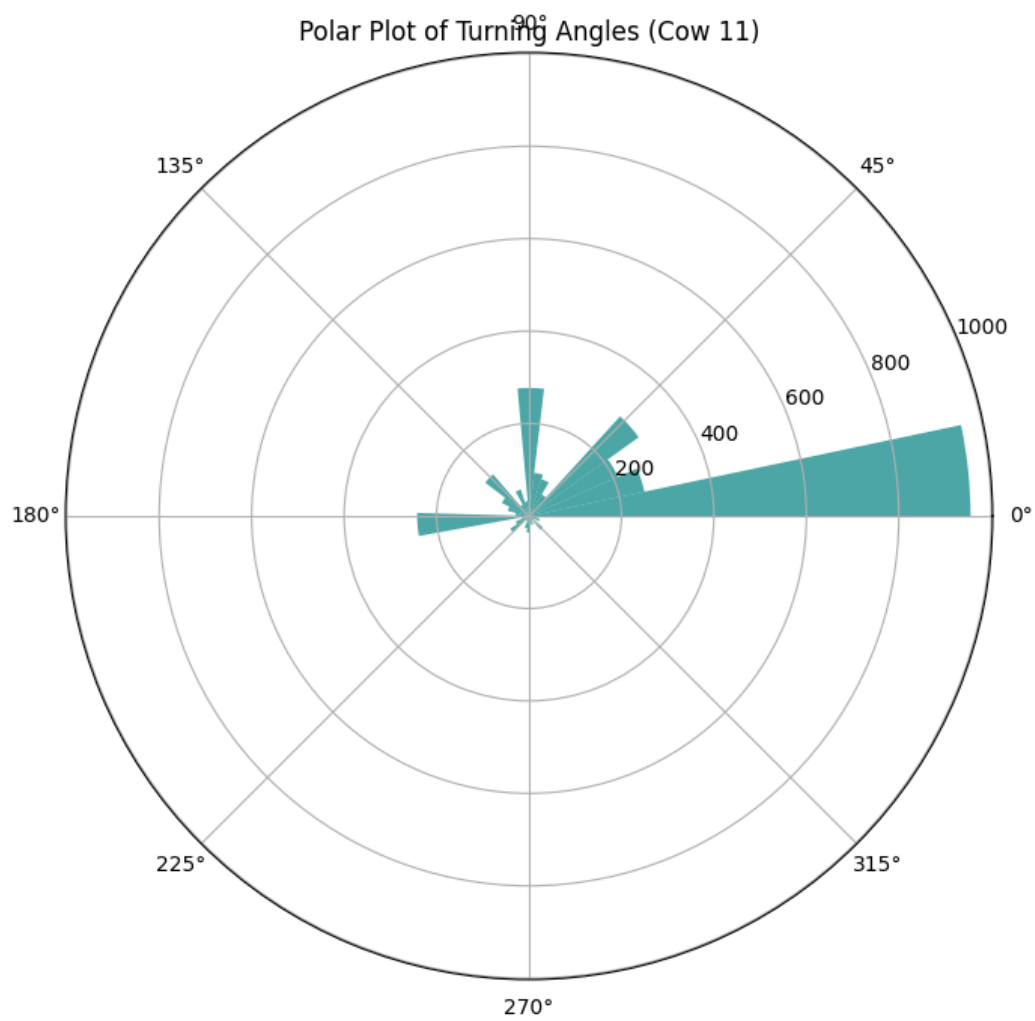


Figure B.23: polar\_turning\_angles\_cow\_11

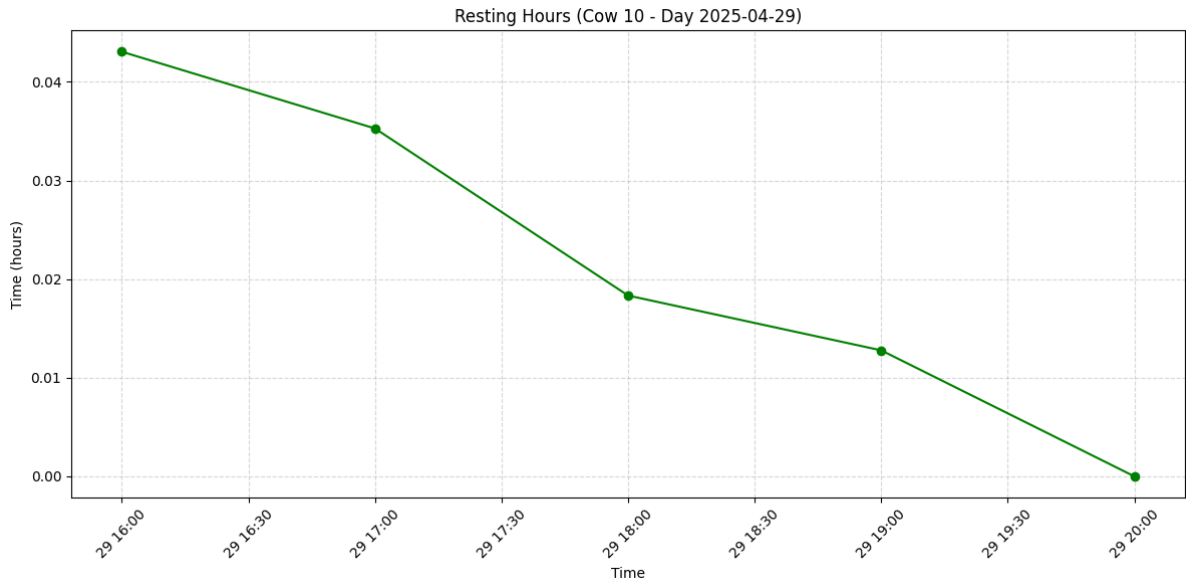


Figure B.24: resting\_hours\_cow\_10\_day

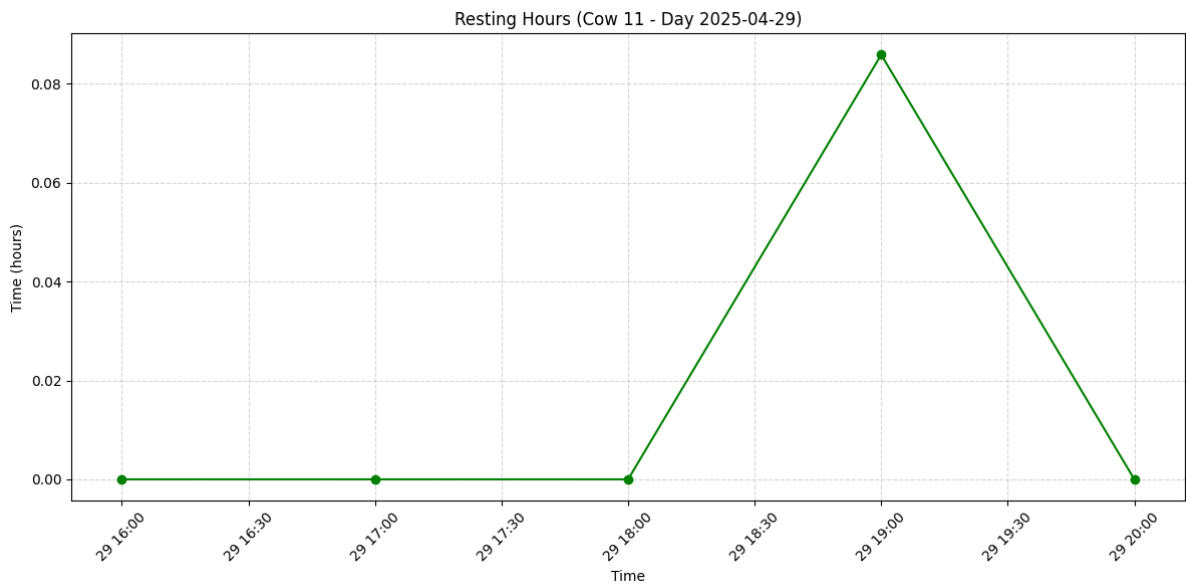


Figure B.25: resting\_hours\_cow\_11\_day

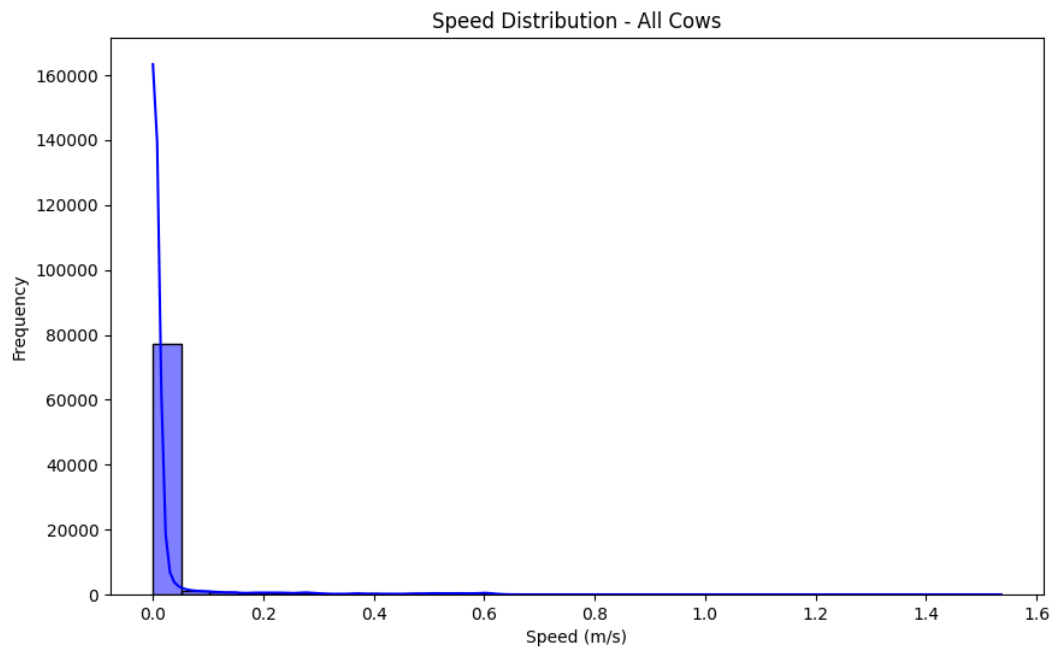


Figure B.26: speed\_distribution\_all

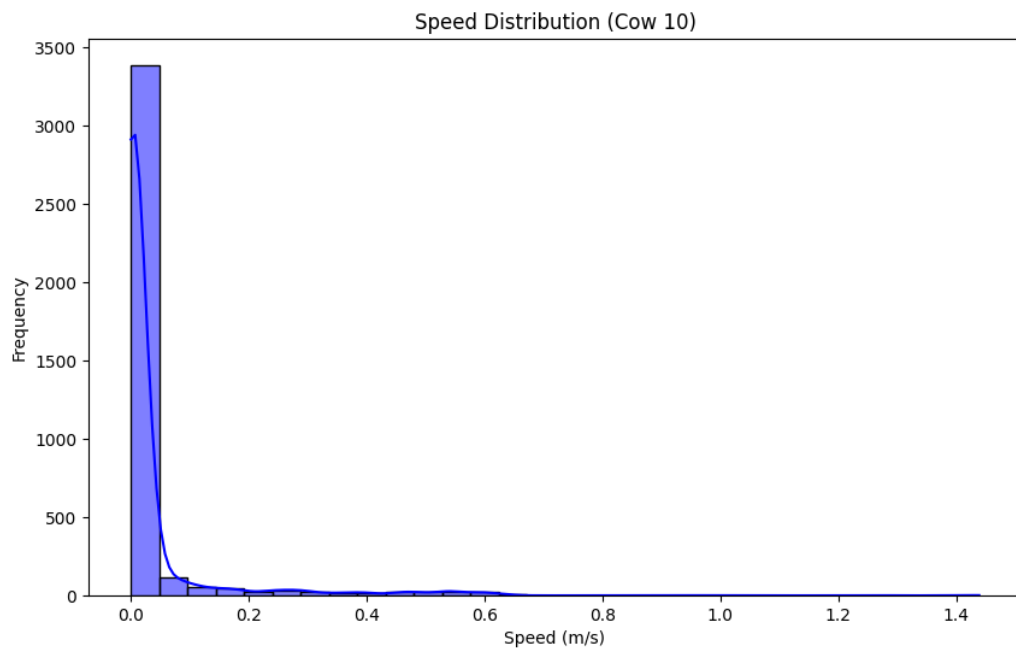


Figure B.27: speed\_distribution\_cow\_10

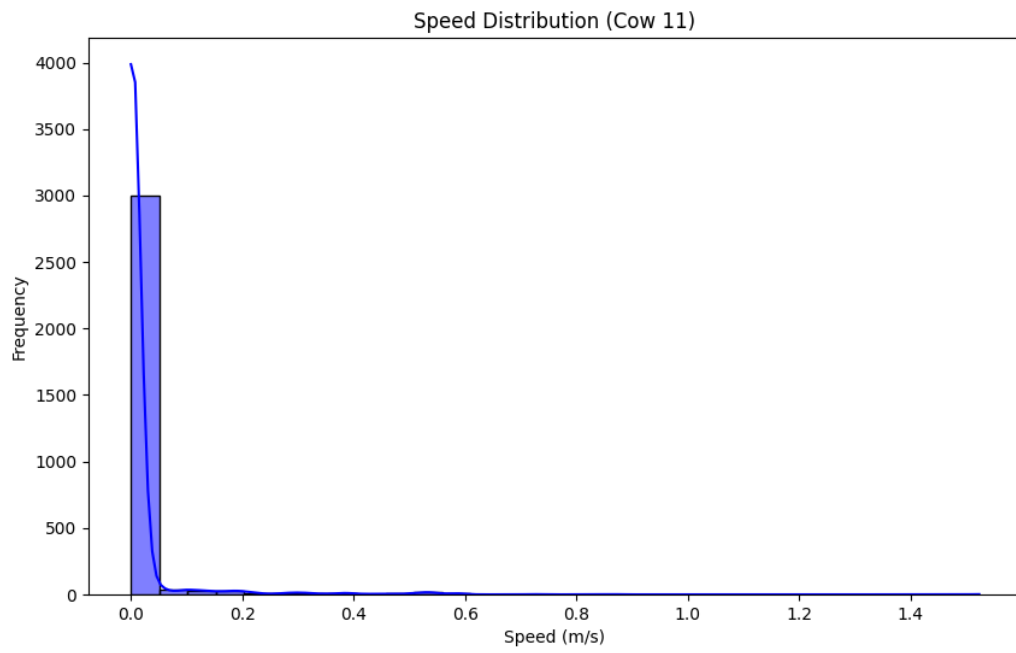


Figure B.28: speed\_distribution\_cow\_11

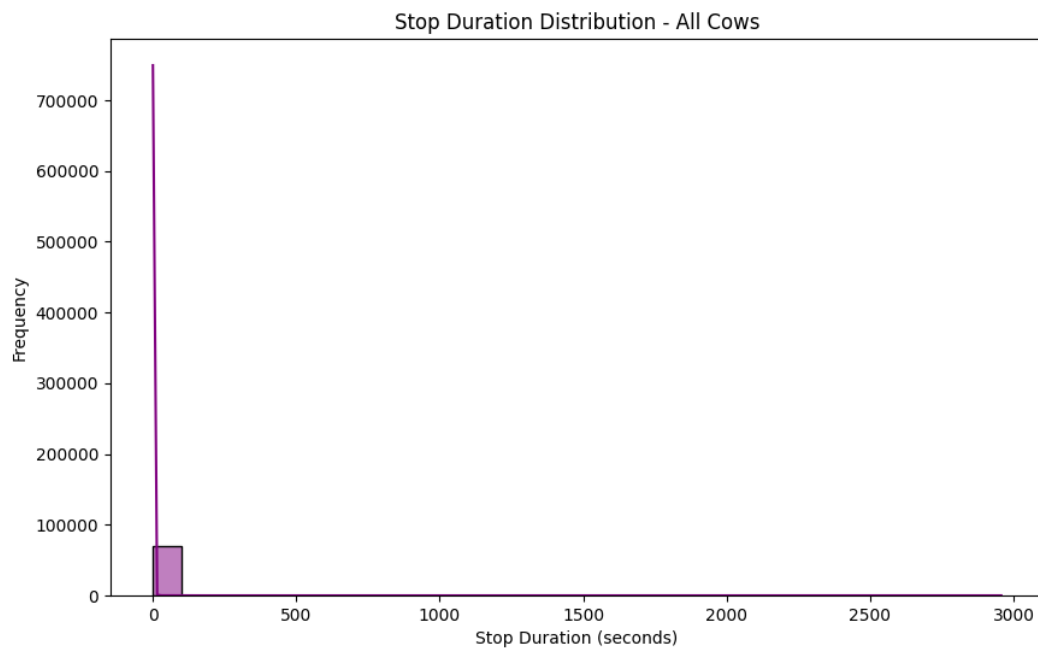


Figure B.29: stop\_duration\_all

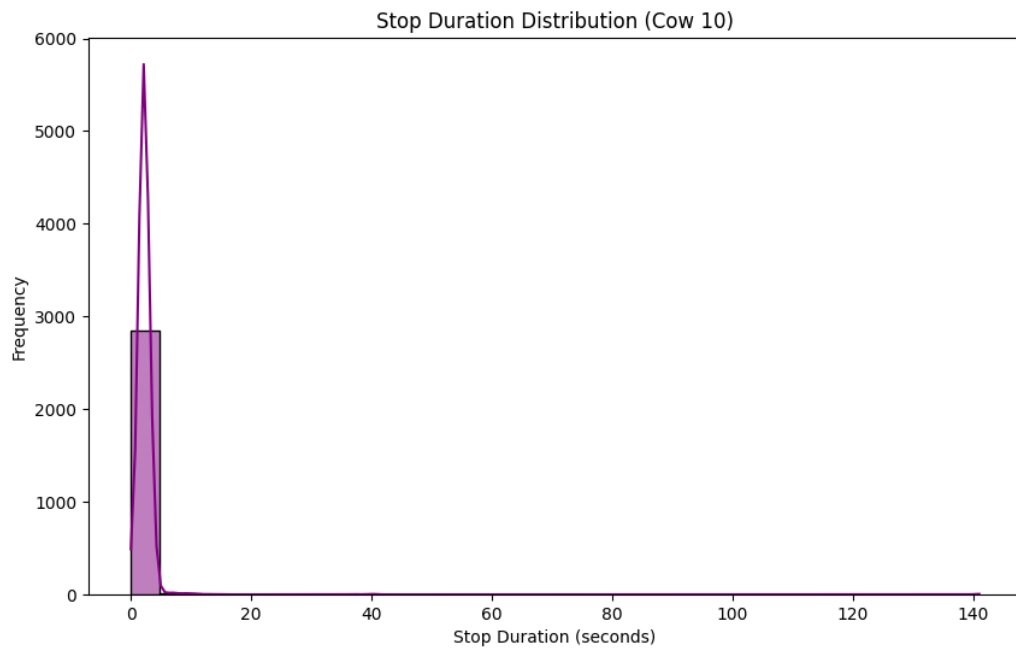


Figure B.30: stop\_duration\_cow\_10

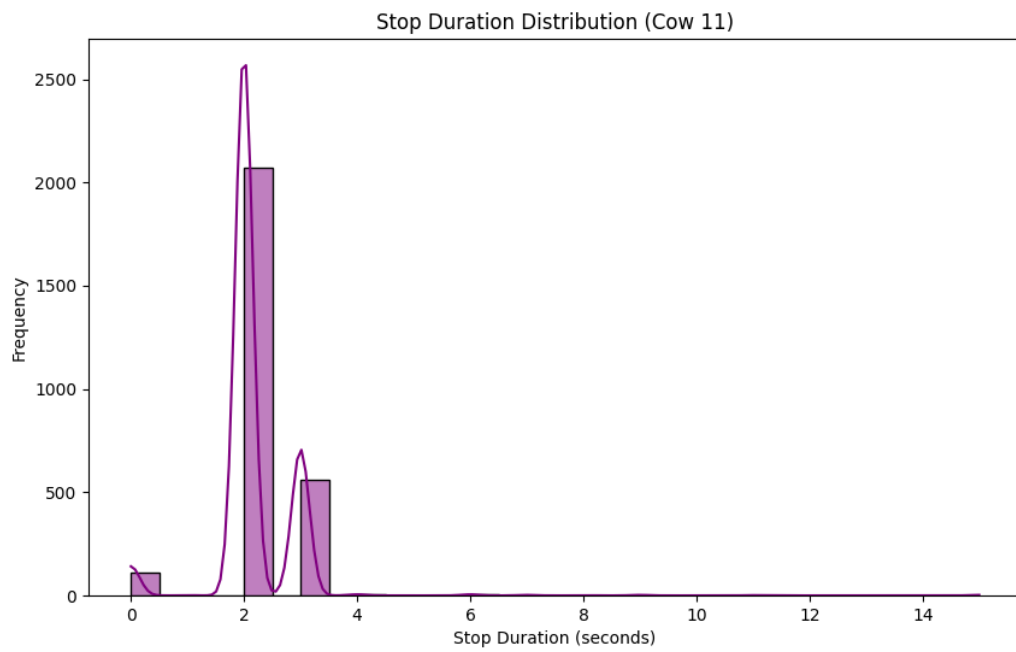


Figure B.31: stop\_duration\_cow\_11

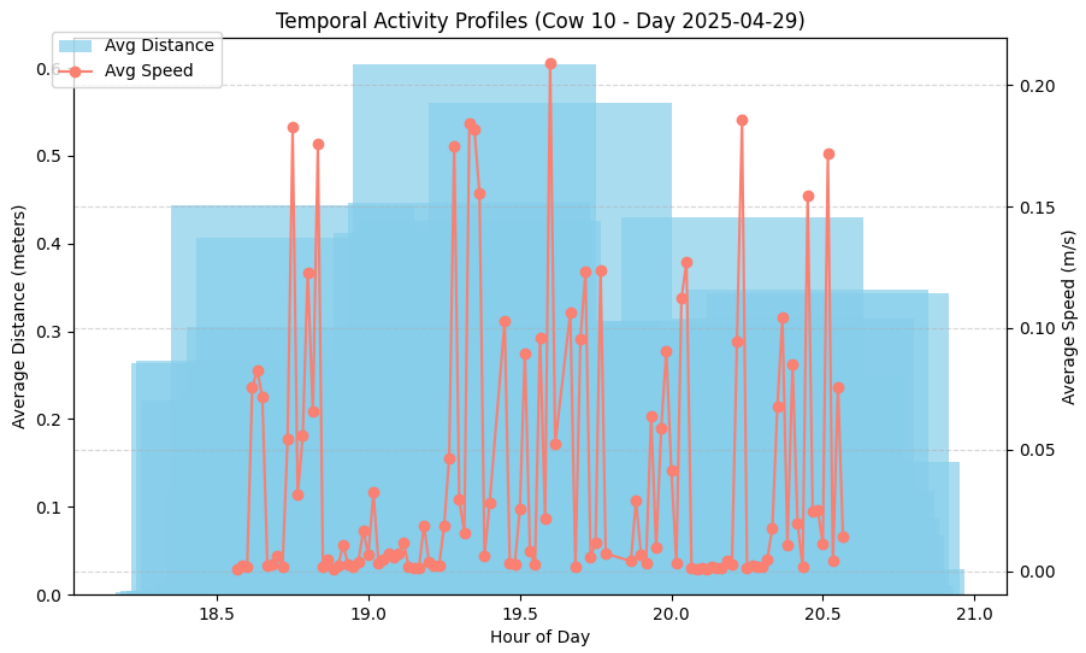


Figure B.32: temp\_activity\_cow\_10\_day

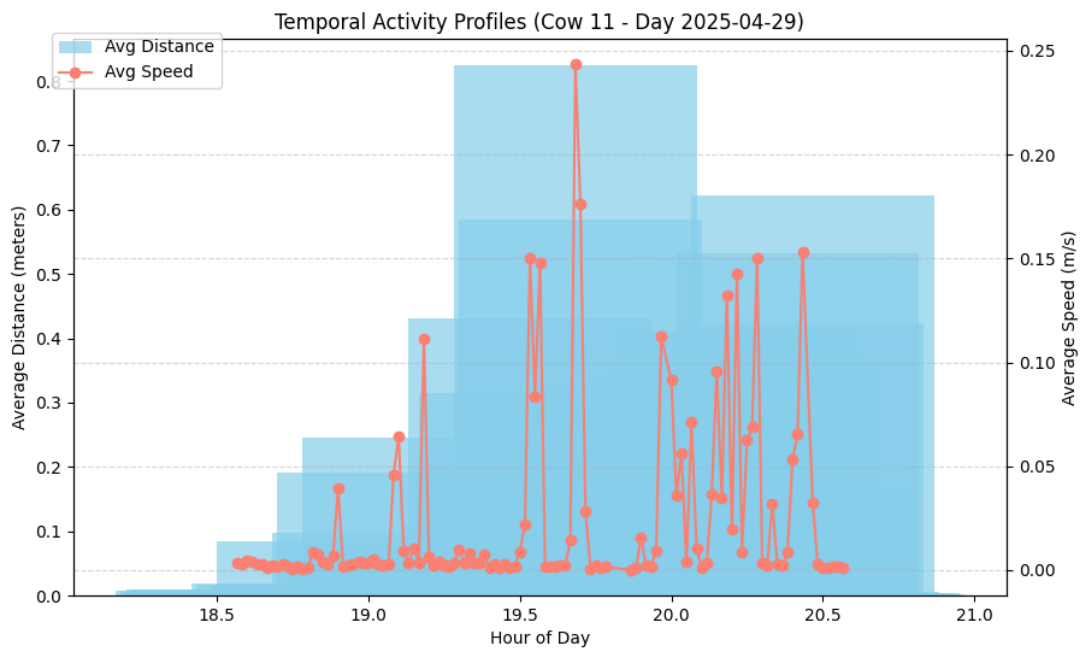


Figure B.33: temp\_activity\_cow\_11\_day

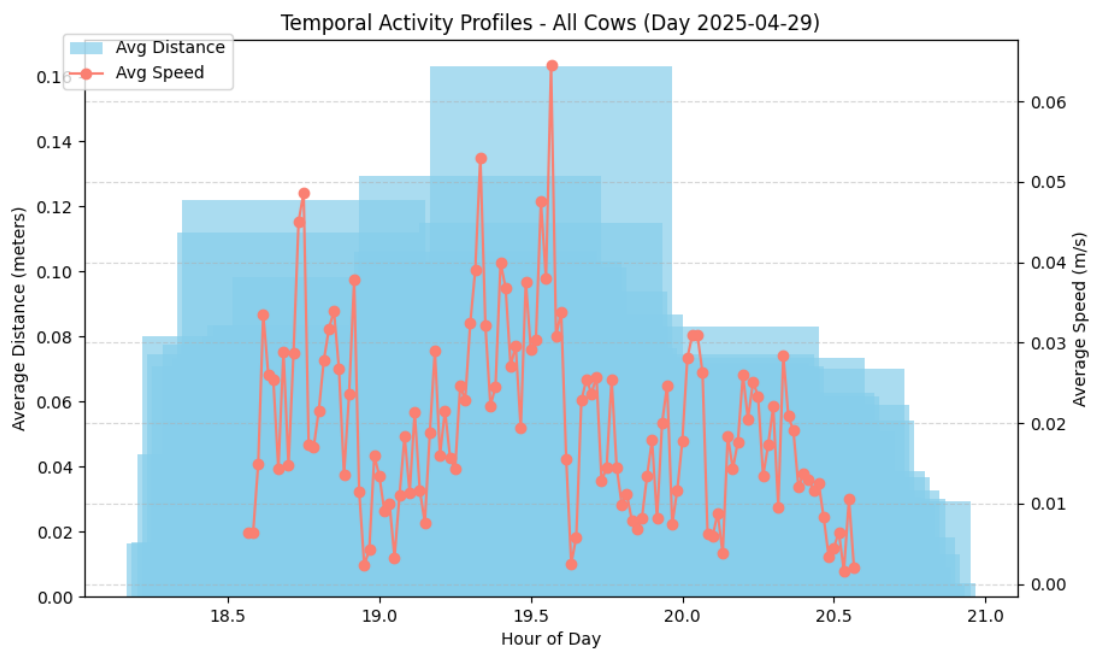


Figure B.34: temp\_activity\_day

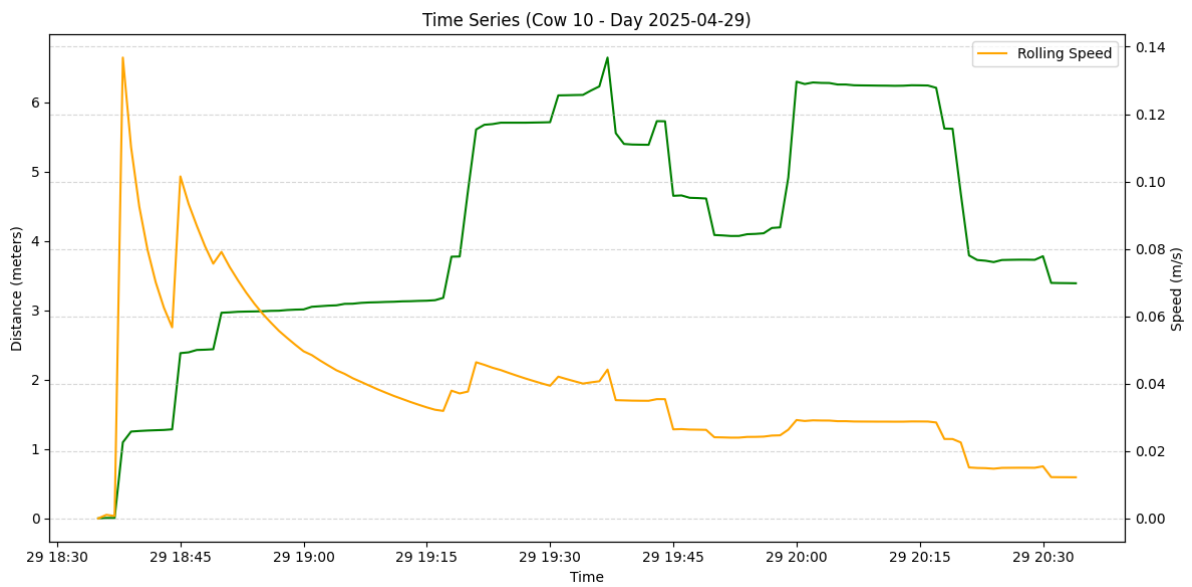


Figure B.35: timeseries\_cow\_10\_day



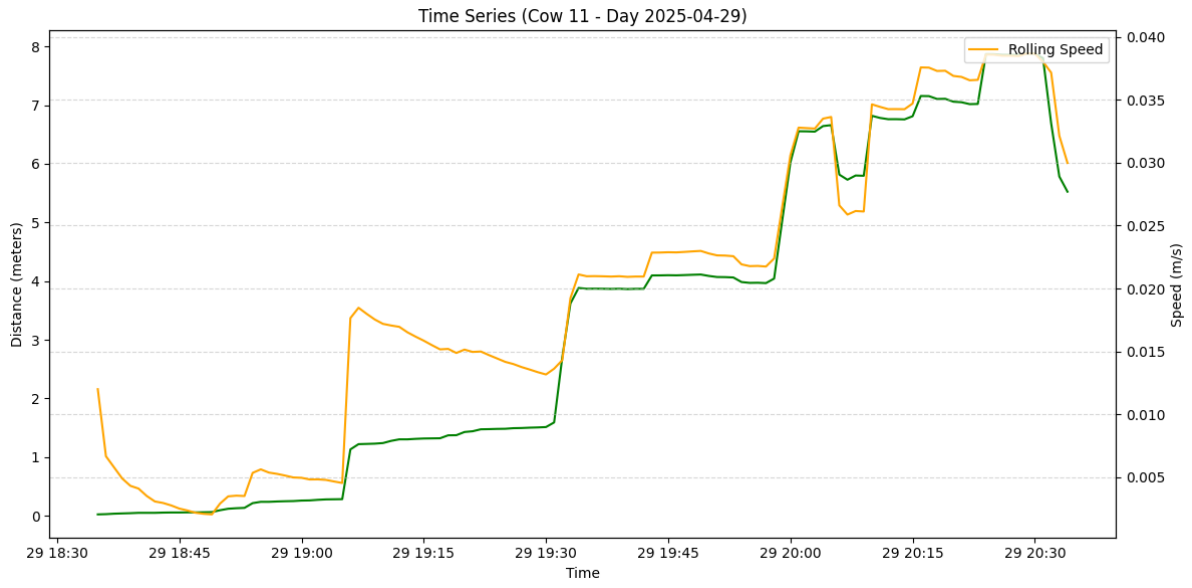


Figure B.36: timeseries\_cow\_11\_day

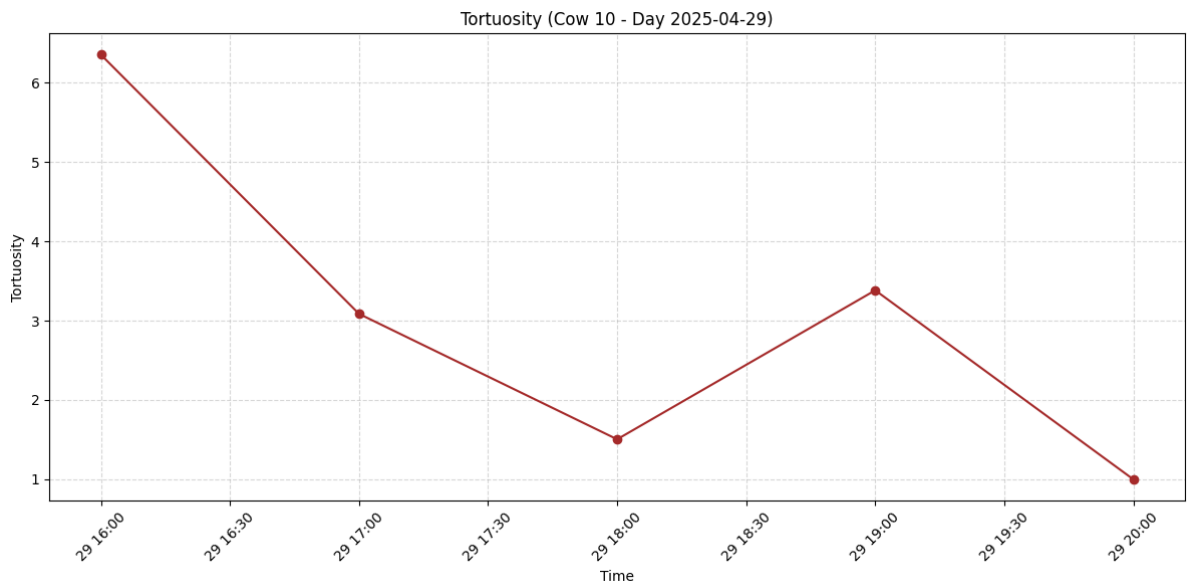


Figure B.37: tortuosity\_cow\_10\_day

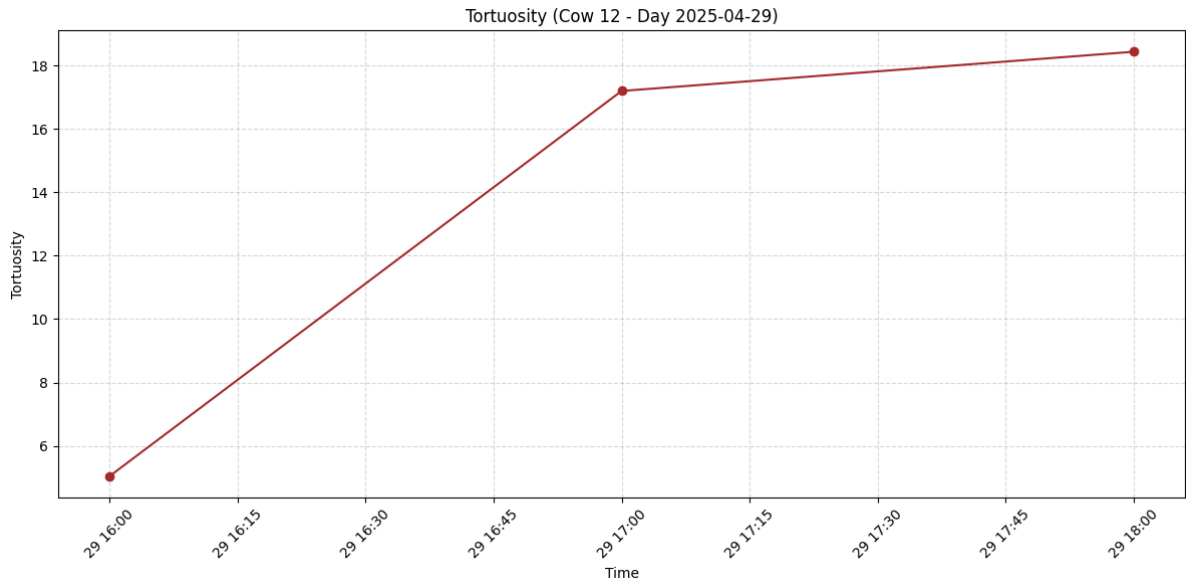


Figure B.38: tortuosity\_cow\_12\_day

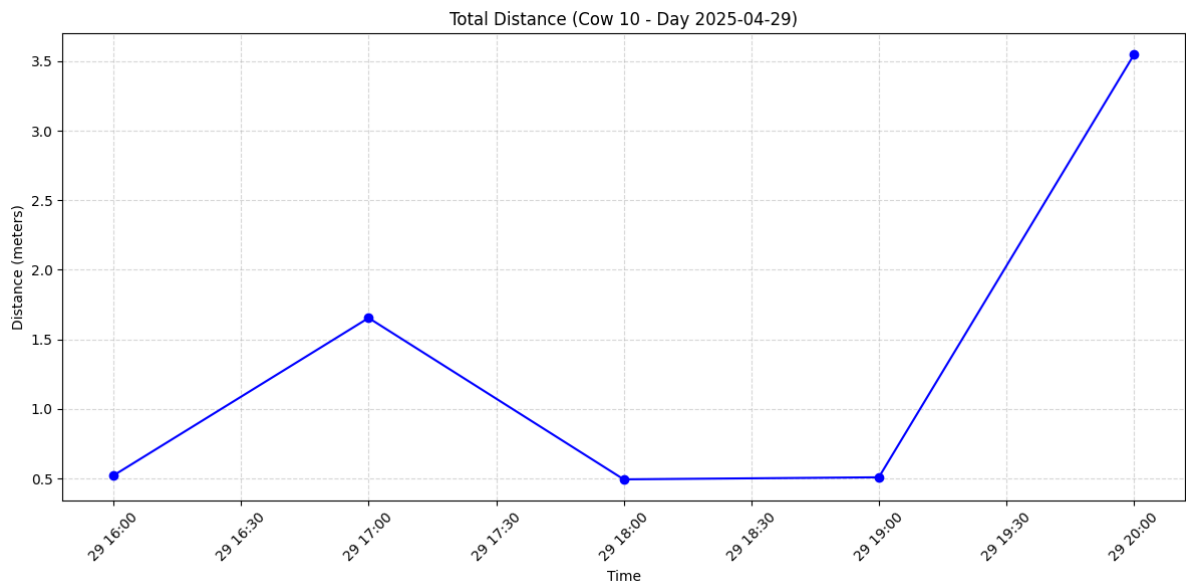


Figure B.39: total\_distance\_cow\_10\_day

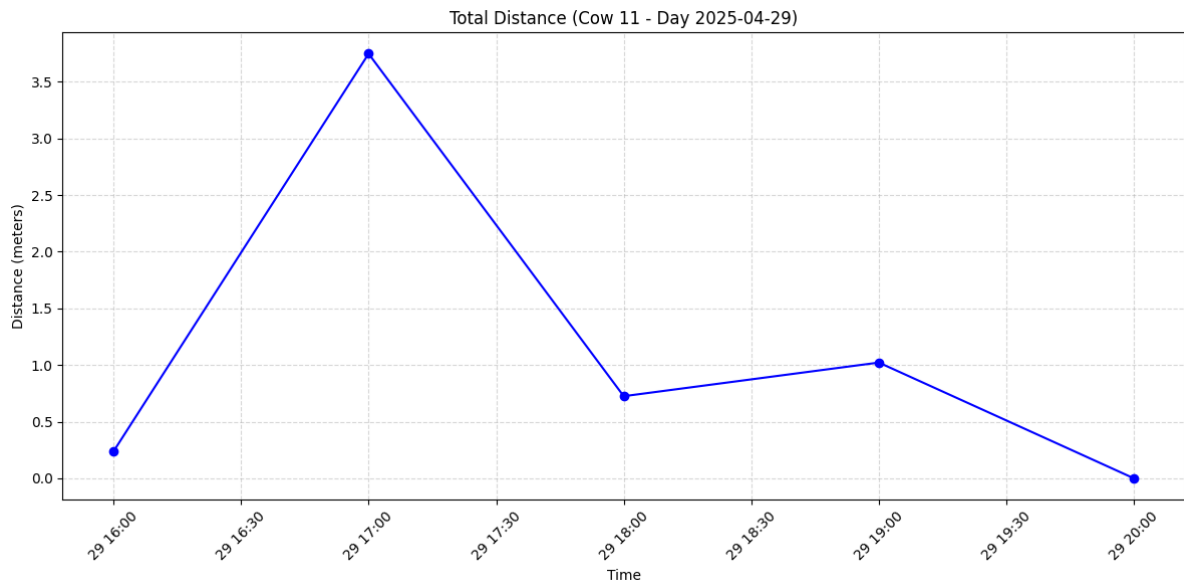


Figure B.40: total\_distance\_cow\_11\_day

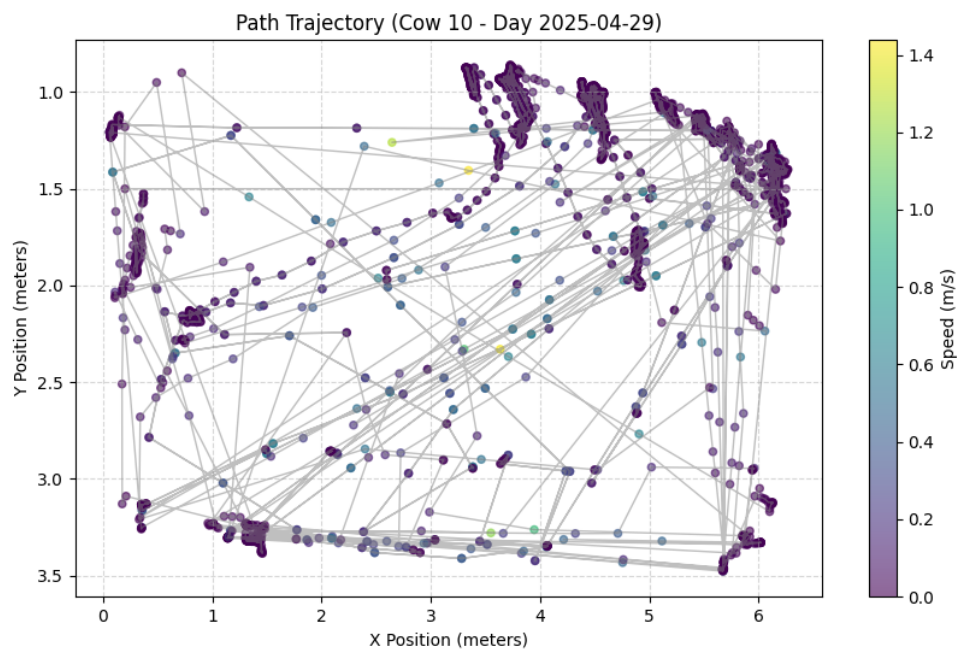


Figure B.41: trajectory\_cow\_10\_day

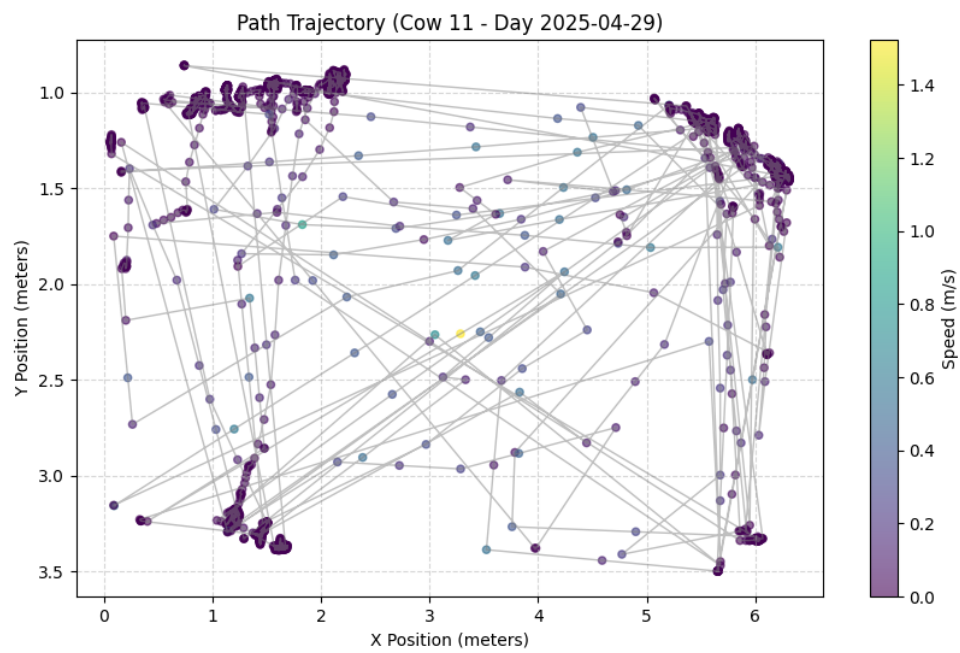


Figure B.42: trajectory\_cow\_11\_day

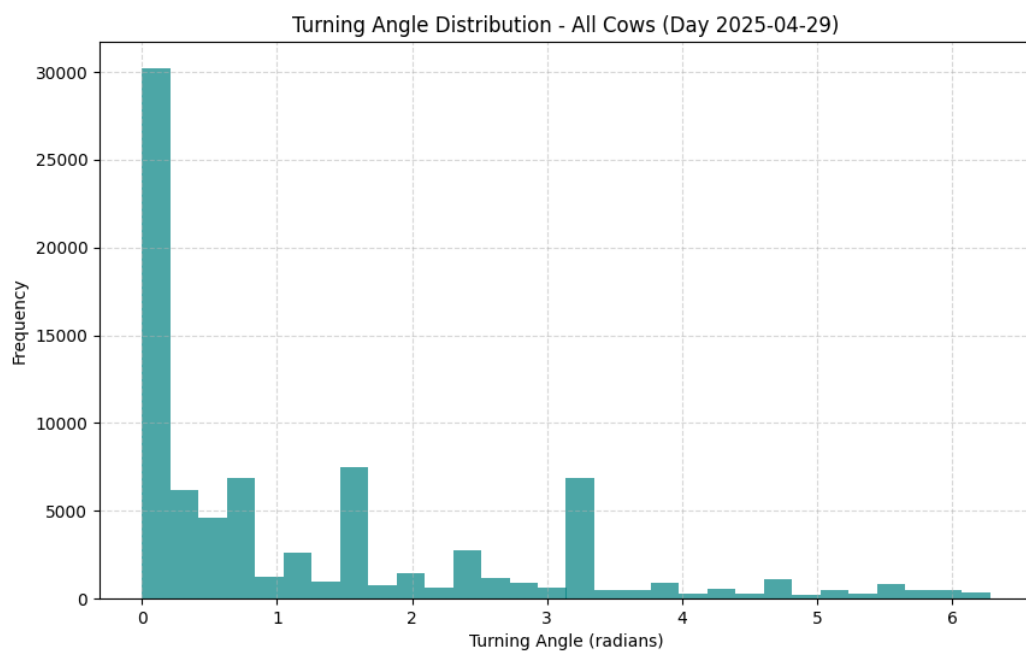


Figure B.43: trajectory\_day

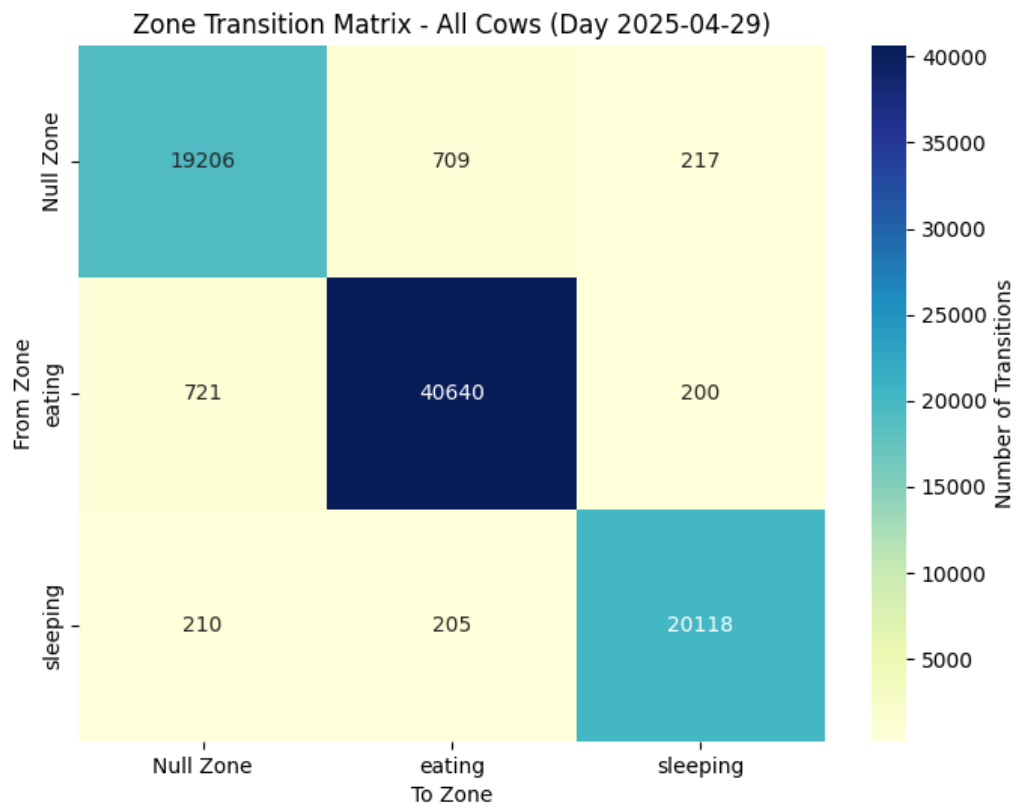


Figure B.44: trans\_matrix\_day

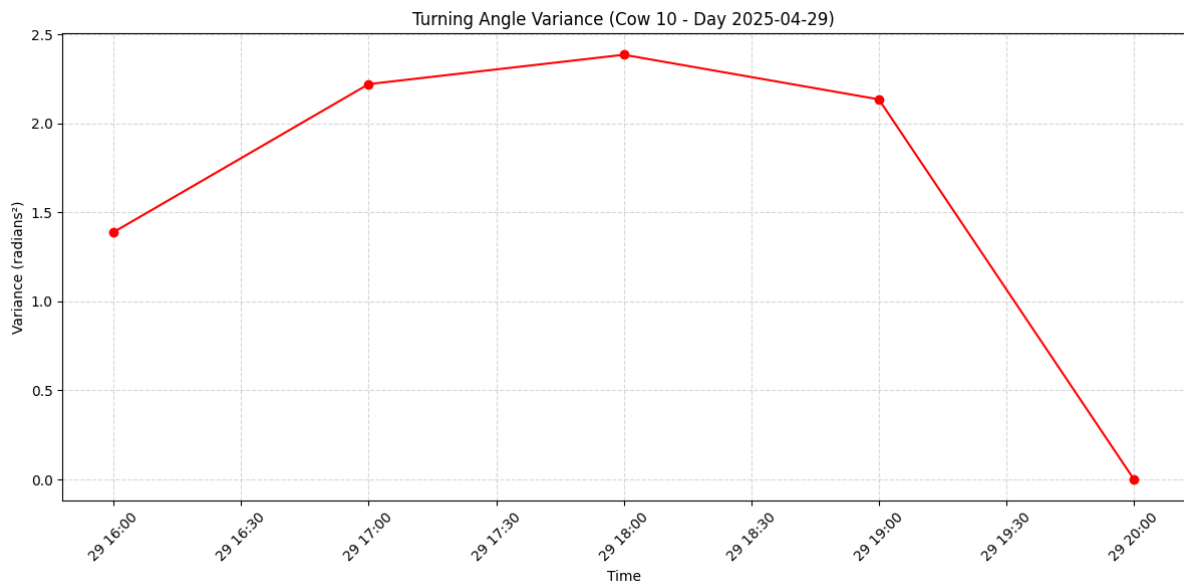


Figure B.45: turning\_variance\_cow\_10\_day

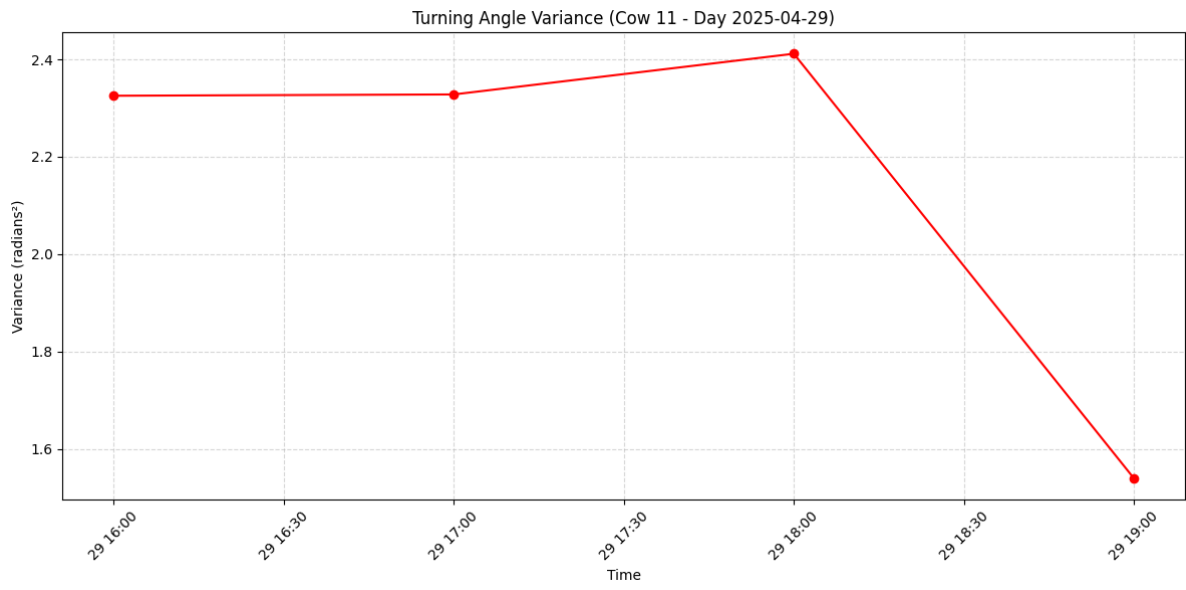


Figure B.46: turning\_variance\_cow\_11\_day

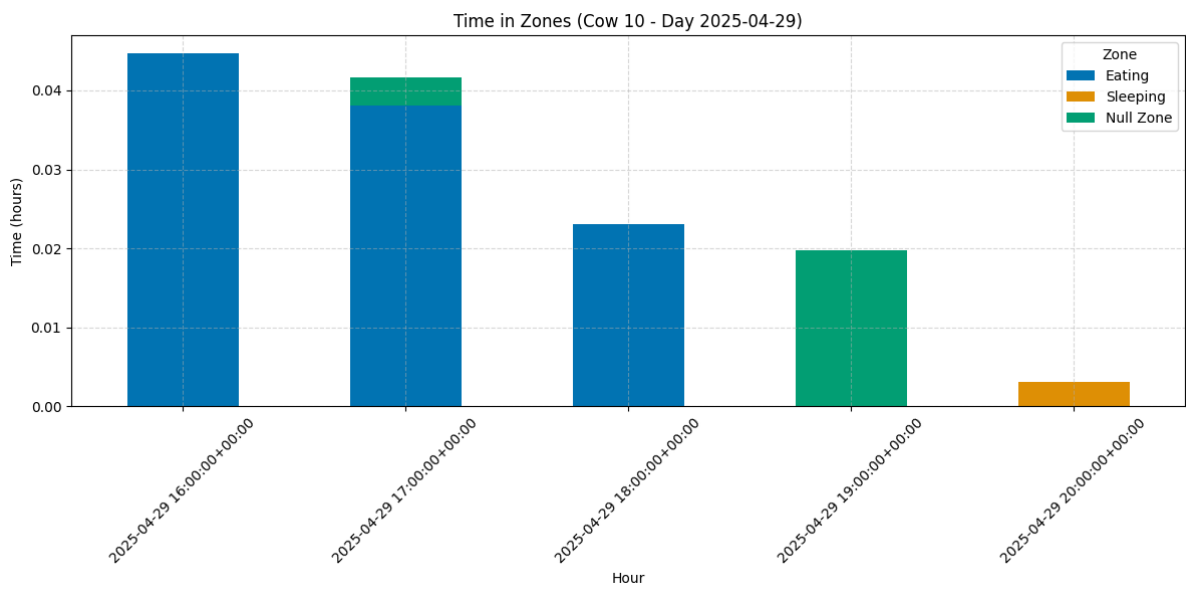


Figure B.47: zone\_time\_cow\_10\_day

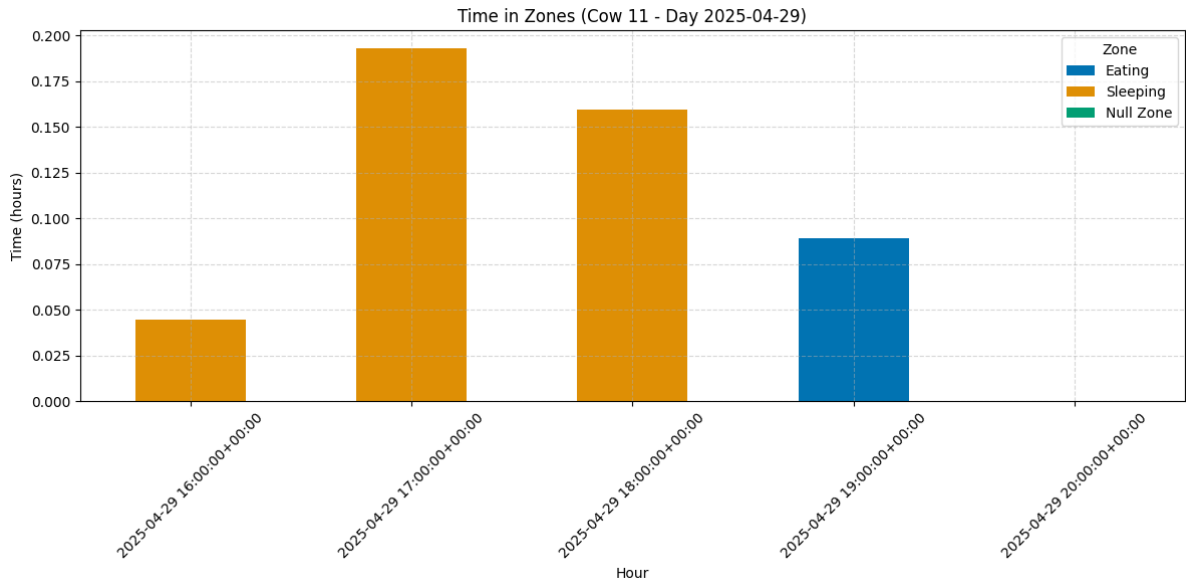


Figure B.48: zone\_time\_cow\_11\_day

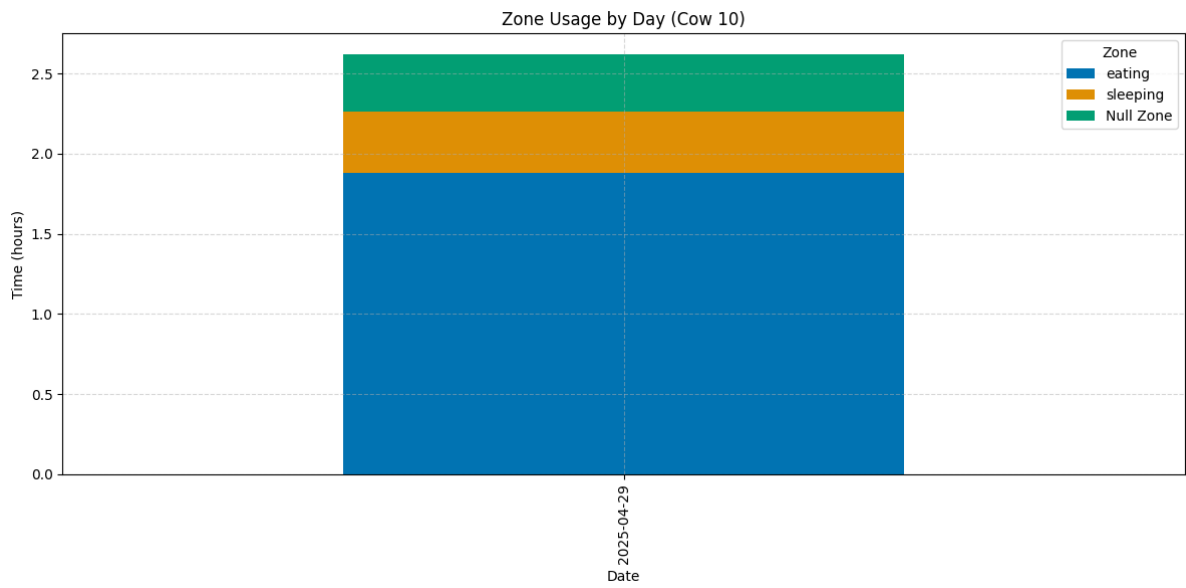


Figure B.49: zone\_usage\_cow\_10

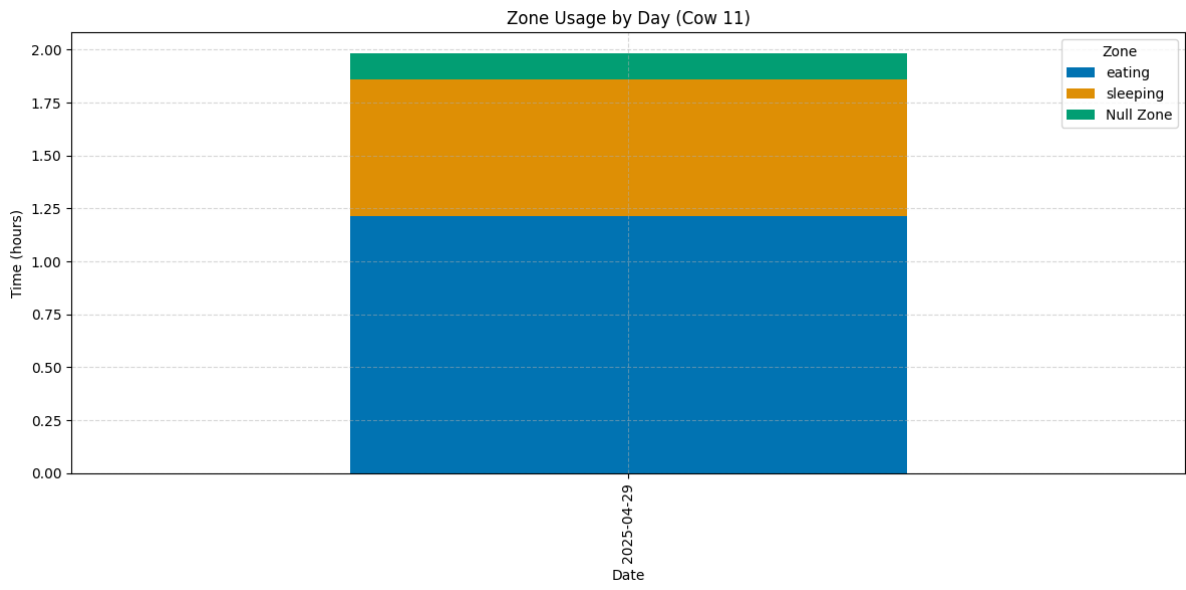


Figure B.50: zone\_usage\_cow\_11